Faculty of Engineering and Technology Electrical and Computer Engineering Department

Database Systems

COMP 333

Database Project

Furniture Showroom Management System

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• Chapter One:

1.1 Introduction

The Furniture Showroom Management System is a comprehensive database-driven application designed to automate and manage the daily operations of a furniture showroom. Traditional furniture showrooms often rely on manual procedures or separate systems to handle sales, inventory, warehouse operations, customer management, and product deliveries. Such approaches may lead to data inconsistency, communication gaps between departments, inventory inaccuracies, delayed deliveries, and reduced customer satisfaction.

This project aims to provide an integrated and centralized solution that efficiently manages all showroom activities through a relational database management system. The system supports different business processes, including customer management, product management, sales processing, inventory control, warehouse management, supplier management, purchase transactions, payment processing, and delivery management.

The system enables customers to browse products, maintain shopping carts and favorite lists, and place orders. Sales employees are responsible for processing customer orders and coordinating with warehouse personnel to verify product availability. Warehouse managers supervise inventory levels, handle warehouse requests, and monitor stock movements, while delivery employees manage delivery operations and update delivery statuses throughout the delivery lifecycle.

The database serves as the core component of the system by storing and managing information related to customers, employees, products, suppliers, warehouses, sales transactions, payments, deliveries, inventory records, purchase transactions, and other operational activities. All data is stored in a structured and relational manner to ensure data integrity, consistency, and secure access.

The system supports multiple user roles, including customers, employees, and administrators. Each user role is provided with specific permissions and functionalities according to its responsibilities, ensuring secure and efficient system operation.

To achieve an efficient database design, the system was modeled using Entity-Relationship (ER) and Enhanced Entity-Relationship (EER) diagrams, which were later transformed into a relational schema and normalized to eliminate redundancy and maintain data consistency. The application was implemented using Java and JavaFX as the application layer and MySQL as the database management system, following modern software engineering principles such as layered architecture and the Model–View–Controller (MVC) architectural pattern.

Overall, the Furniture Showroom Management System provides an integrated environment for managing showroom operations, improving communication between departments, reducing manual work, enhancing operational efficiency, and increasing customer satisfaction.

1.2 Problem Statement

Many furniture showrooms still rely on manual procedures or fragmented software systems to manage their daily operations. These traditional approaches often make it difficult to efficiently manage customer orders, monitor inventory levels, coordinate warehouse activities, and track product deliveries.

The absence of an integrated management system may result in several operational problems, including inventory inconsistencies, delayed order processing, communication gaps between sales and warehouse departments, inaccurate stock information, and difficulties in monitoring delivery operations. In addition, manually managing customer information, supplier records, payments, and purchase transactions increases the likelihood of human errors and data redundancy.

Another major challenge is the lack of real-time information sharing among different departments. For example, sales employees may approve customer orders without having accurate information about product availability in warehouses, leading to customer dissatisfaction and operational delays.

Therefore, there is a need for a centralized and integrated system capable of managing all showroom activities through a single platform. The proposed Furniture Showroom Management System aims to address these challenges by providing efficient data management, improving communication between departments, automating business processes, and ensuring accurate and consistent information throughout the system.

The main objective of the Furniture Showroom Management System is to develop an integrated database application that automates and manages the daily operations of a furniture showroom efficiently and accurately.

1.3 Project Objectives

The specific objectives of the project are as follows:

- To provide a centralized system for managing all showroom activities through a single platform.
- To manage customer information, employee records, suppliers, and product data efficiently.
- To facilitate sales processing and customer order management.
- To improve inventory control and warehouse management by providing accurate and real-time stock information.
- To enhance communication and coordination between sales employees and warehouse personnel.
- To support delivery management and enable tracking of delivery operations and statuses.
- To manage supplier information, purchase transactions, and stock replenishment processes.
- To reduce manual work, minimize human errors, and eliminate data redundancy.

- To ensure data consistency, integrity, and secure access through a relational database management system.
- To provide an easy-to-use graphical user interface that allows users to interact with the system efficiently.
- To generate a scalable and maintainable system that can support future enhancements and additional functionalities.

1.4 System Overview

The Furniture Showroom Management System is a database-driven application designed to support and automate the core business activities of a furniture showroom. The system integrates multiple functional modules into a single platform to ensure efficient data management and seamless communication among different departments.

The system consists of several major modules, including Customer Management, Product Management, Sales Management, Warehouse Management, Inventory Management, Supplier Management, Purchase Management, Delivery Management, Payment Management, and Reporting. These modules work together to streamline business operations and improve overall showroom performance.

The system supports multiple categories of users, each with specific responsibilities and permissions. Customers can browse available products, manage shopping carts and favorite lists, and place orders. Employees, depending on their roles, can process sales transactions, manage warehouse requests, monitor inventory levels, supervise delivery operations, and maintain supplier and purchase information. Administrators are responsible for overseeing system operations and managing user accounts and system data.

The application follows a layered architecture combined with the Model–View–Controller (MVC) design pattern to ensure modularity, maintainability, and scalability. Java and JavaFX are used to implement the application layer, while MySQL serves as the underlying database management system.

By integrating all showroom activities into a unified environment, the system improves operational efficiency, reduces manual processing, facilitates interdepartmental communication, and enhances customer service.

- # Chapter Two:

2.1 Functional Requirements

Functional requirements describe the main services and operations provided by the Furniture Showroom Management System. These requirements define the interactions between users and the system and specify the functionalities that must be implemented to support the daily operations of the furniture showroom.

The system is designed to support multiple user categories, including customers, employees, and administrators. Each user category is provided with specific functionalities according to its responsibilities and access privileges.

The following sections describe the functional requirements of each user category as well as the core functions performed automatically by the system.

2.1.1 Customer Functions

Customers represent one of the primary user categories in the Furniture Showroom Management System. The system provides customers with several functionalities that enable them to interact with the showroom efficiently and conveniently.

Customers can perform the following functions:

- Create a new account and securely log into the system.
- Manage and update their personal account information.
- Browse and search available furniture products.
- Filter products according to different criteria such as category, color, material, and availability status.
- View detailed information about products, including descriptions, prices, and images.
- Add products to their shopping cart and manage cart contents by updating quantities or removing items.
- Add products to a favorites list for future reference.
- Place new orders by selecting products from the shopping cart.
- View current and previous orders along with their statuses.
- Cancel an order only if its status is still pending and has not yet entered the processing stage.
- Track order progress and monitor delivery status.
- Submit product return requests for purchased products according to the showroom return policy.

These functionalities provide customers with a convenient shopping experience while ensuring efficient communication with the showroom and accurate management of customer transactions.

2.1.2 Employee Functions

The Furniture Showroom Management System supports multiple employee roles, each responsible for specific operational tasks within the showroom. The system provides dedicated functionalities for each employee category according to its responsibilities.

- ### Sales Employee Functions

Sales employees are responsible for managing customer orders and coordinating with other departments. Their main functions include:

- Create and process customer orders.
- Review and approve customer sales transactions.
- Monitor order statuses and update order information.
- Communicate with warehouse personnel by sending warehouse requests to verify product availability.
- Review warehouse responses and make appropriate decisions regarding customer orders.
- Monitor sales performance and daily sales activities.
- View recent activities, alerts, and pending orders.

- ### Warehouse Manager Functions

Warehouse managers are responsible for supervising warehouse operations and inventory management. Their functions include:

- Monitor inventory levels and warehouse stock.
- Review and respond to warehouse requests submitted by sales employees.
- Check product availability within warehouses.
- Approve, reject, or partially approve warehouse requests.
- Manage warehouse inventory and stock quantities.
- Monitor stock movements and warehouse activities.
- View warehouse performance statistics and alerts.

- ### Delivery Employee Functions

Delivery employees are responsible for handling product deliveries and updating delivery operations. Their functions include:

- View assigned delivery tasks.
- Manage delivery schedules and assigned orders.
- Update delivery statuses such as pending, in progress, delivered, or cancelled.
- Record delivery notes and report delivery issues when necessary.
- Monitor delivery performance and completed deliveries.
- View recent delivery activities and assigned tasks.

2.1.3 Administrator Functions

The administrator is responsible for supervising and managing the overall operation of the Furniture Showroom Management System. The administrator is granted full access privileges to ensure efficient system management and data maintenance.

The administrator can perform the following functions:

- Manage customer and employee accounts.
- Monitor and maintain system data.
- Add, update, or remove products from the system.
- Manage product categories and supplier information.
- Manage warehouse information and inventory records.
- Monitor sales transactions, purchase transactions, and payment records.
- View and supervise delivery operations and order statuses.
- Access reports, statistics, and system performance indicators.
- Monitor user activities and ensure proper system operation.
- Maintain data consistency and support system administration tasks.

The administrator plays a critical role in ensuring that all system modules operate correctly and that the stored information remains accurate, consistent, and up to date.

2.1.4 System Functions

In addition to user-specific functionalities, the Furniture Showroom Management System performs several automated operations to ensure efficient and accurate system behavior. These functions are executed automatically by the system without direct user intervention.

The system performs the following functions:

- Authenticate users and verify login credentials before granting access to system resources.
- Manage user access permissions according to assigned roles and responsibilities.
- Store, retrieve, update, and maintain data related to customers, employees, products, suppliers, warehouses, and transactions.
- Validate user input and prevent invalid or incomplete data from being stored in the database.
- Automatically update order statuses throughout the sales and delivery lifecycle.
- Facilitate communication between sales employees and warehouse managers through warehouse requests and responses.
- Maintain inventory consistency by updating stock quantities after sales, purchases, and product returns.
- Record stock movement activities for inventory tracking purposes.
- Manage relationships between different system modules and ensure data consistency across all operations.

- Generate performance indicators, statistics, alerts, and reports to support decision-making.
- Ensure data integrity by enforcing database constraints such as primary keys, foreign keys, and business rules.

These automated functions contribute to improving operational efficiency, reducing manual errors, and ensuring reliable system performance.

2.2 Non-Functional Requirements

Non-functional requirements describe the quality attributes and operational characteristics of the Furniture Showroom Management System. These requirements ensure that the system operates efficiently, securely, and reliably while providing a satisfactory user experience.

Performance

The system should provide fast and efficient execution of operations such as product searching, order processing, inventory updates, and data retrieval. Most system operations should be completed within a few seconds to ensure smooth interaction, even when handling large amounts of data.

Usability

The system should provide a user-friendly and intuitive graphical user interface that enables users to navigate easily through different modules. The interfaces should be consistent, visually organized, and easy to understand, allowing users with different technical backgrounds to use the system efficiently.

Reliability

The system should operate consistently and accurately without unexpected failures. Data transactions and system operations must be executed correctly to ensure dependable system behavior and minimize operational errors.

Security

The system should ensure secure access by authenticating users before granting access to system resources. Access permissions should be controlled according to user roles to prevent unauthorized access. Additionally, input validation and secure database operations should be implemented to maintain data security and integrity.

Maintainability

The system should be designed using a modular architecture to simplify future maintenance, debugging, and feature enhancements. The use of design patterns such as MVC and the separation of system layers facilitate system modifications and improvements.

Scalability

The system should support future expansion by allowing new functionalities, additional users, and new business modules to be integrated without requiring major modifications to the existing system structure.

2.3 Domain Description

The proposed system belongs to the retail management and inventory management domain. It is specifically designed to support the operations of furniture showrooms by integrating sales, inventory, warehouse, supplier, and delivery management into a unified system.

Furniture showrooms typically deal with a large number of products, customers, suppliers, and business transactions on a daily basis. Managing these activities manually or through disconnected systems may lead to inventory inaccuracies, delayed order processing, communication problems between departments, and inefficient business operations.

The Furniture Showroom Management System addresses these challenges by providing a centralized platform that automates and coordinates the main business processes within the showroom. The system facilitates communication between sales employees, warehouse managers, delivery personnel, and administrators while ensuring accurate and consistent data management.

The system supports various operational activities, including customer order processing, inventory monitoring, warehouse management, supplier management, purchase transactions, delivery tracking, and payment processing. By integrating these activities into a single environment, the system contributes to improving operational efficiency, enhancing customer service, and supporting effective business decision-making.

Overall, the proposed system represents a comprehensive solution for modern furniture showroom management, supporting business growth and improving the utilization of organizational resources through efficient database and application design.

• Chapter Three: System Architecture and Technologies

3.1 Introduction

The Furniture Showroom Management System was designed using a structured architecture to support the various business operations of the showroom efficiently. Since the system integrates multiple functional modules such as customer management, product management, sales processing, warehouse operations, inventory control, purchasing, payments, and delivery management, a well-organized architectural design was required to ensure smooth communication between system components and reliable data management.

To achieve maintainability, scalability, and separation of concerns, the system was implemented using Java and JavaFX as the application layer and MySQL as the database management system. In addition, the application adopts a layered architecture and the Model–View–Controller (MVC) design pattern to separate presentation, business logic, and data access responsibilities.

The following sections describe the technologies used in the system and the architectural views adopted during the design and implementation process.

3.2 Technologies Used

Several technologies and tools were utilized during the development of the Furniture Showroom Management System. These technologies were selected to ensure efficient system implementation, reliable database management, and an interactive graphical user interface.

Table 1: Technologies Used in the Furniture Showroom Management System

Technology / Tool	Purpose
Java	Core programming language used for implementing the application logic and system functionalities.
JavaFX	Used for developing the graphical user interface (GUI) of the application.
FXML	Used to define and organize user interface layouts in a structured manner.
CSS	Used for styling and customizing the appearance of user interfaces.
MySQL	Relational Database Management System used for storing and managing system data.
JDBC	Used to establish connectivity between the Java application and the MySQL database.

Scene Builder	Used for visually designing and editing JavaFX interfaces.
Git and GitHub	Used for source code management and version control during the development process.

3.3 Layered Architecture

The Furniture Showroom Management System follows a layered architecture to ensure modularity, maintainability, and separation of concerns. This architectural approach divides the system into several layers, where each layer is responsible for a specific set of functionalities.

The Presentation Layer represents the user interface of the application and is implemented using JavaFX, FXML, and CSS. This layer allows users to interact with the system through graphical interfaces.

The Business Logic Layer contains the application controllers responsible for processing user requests, validating input, and coordinating interactions between the user interface and the database layer.

The Data Access Layer consists of DAO (Data Access Object) classes that manage database operations such as inserting, updating, deleting, and retrieving data from the database.

The MVC pattern operates inside the layered architecture, where the View belongs to the Presentation Layer, Controllers belong to the Business Logic Layer, and Models/DAO classes interact with the Data Access Layer.

Finally, the Database Layer is implemented using MySQL and serves as the central repository for storing and managing all system data.

This layered architecture improves system organization, simplifies maintenance, and facilitates future system enhancements.

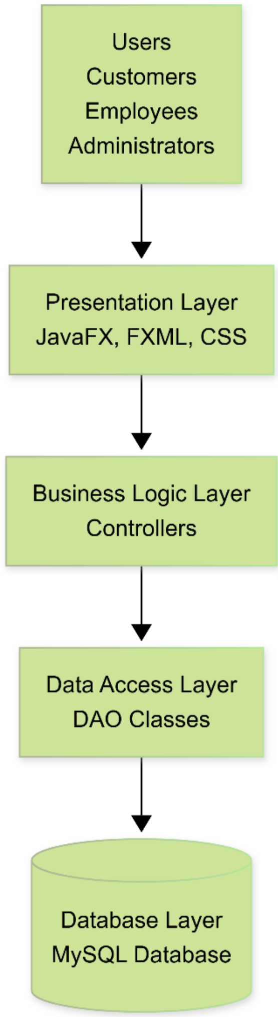


Figure 1 : Layered Architecture of the Furniture Showroom Management System

3.4 MVC Architecture

The Furniture Showroom Management System adopts the Model–View–Controller (MVC) architectural pattern to improve system organization and separate application responsibilities. The MVC pattern divides the application into three main components: Model, View, and Controller.

The View component represents the graphical user interfaces developed using JavaFX, FXML, and CSS. It is responsible for presenting data to users and receiving user interactions.

The Controller component acts as an intermediary between the View and the Model. Controllers process user requests, validate input data, execute application logic, and coordinate interactions among different system components.

The Model component represents the application's data structures and business entities, such as Customer, Employee, Product, and Sale. Model classes work together with DAO classes to retrieve and store data in the database.

DAO classes are responsible for performing database operations and communicating directly with the MySQL database. This separation of concerns improves code maintainability, reusability, and scalability.

By adopting the MVC architectural pattern, the system achieves better modularity, simplifies debugging and maintenance tasks, and facilitates future enhancements.

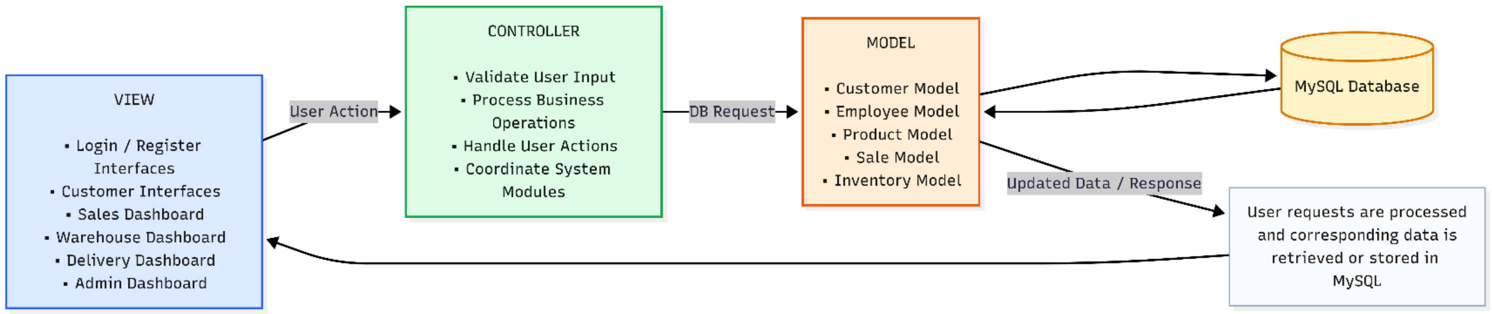


Figure 2 :MVC Architecture of the Furniture Showroom Management System

3.5 Database Connectivity Architecture

The Furniture Showroom Management System follows a client–server architecture in which the JavaFX application acts as the client, while the MySQL database serves as the database server.

Users, including customers, employees, and administrators, interact with the system through the JavaFX graphical user interface. User requests are processed by the application and then transmitted to the MySQL database through JDBC connectivity.

The MySQL server is responsible for storing, retrieving, and managing all system data, including customer information, products, sales transactions, inventory records, warehouse activities, and delivery information.

This architecture provides centralized data management, ensures data consistency, and allows multiple users to interact with the same database efficiently. Furthermore, the client–server approach improves system scalability, maintainability, and security.

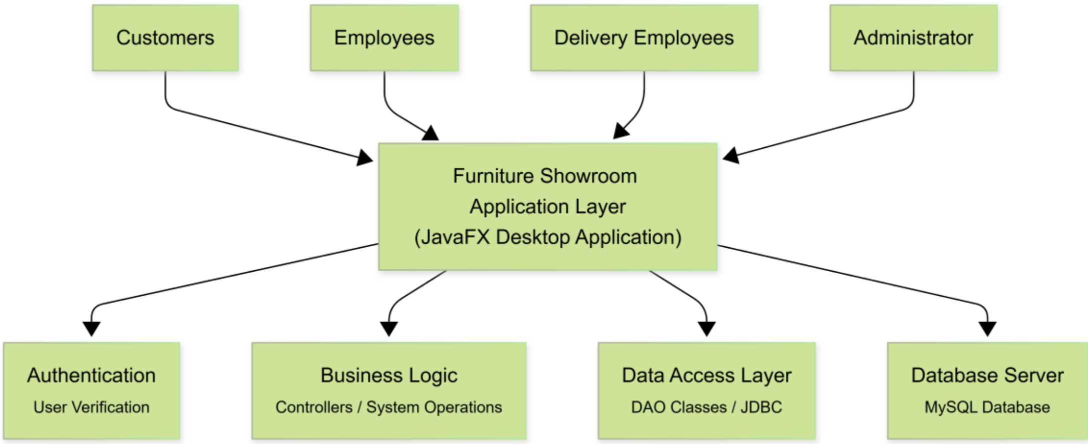


Figure 3: Database Connectivity Architecture of the Furniture Showroom Management System

3.6 Final Integrated Architecture

The Furniture Showroom Management System adopts an integrated architecture that combines layered and MVC architectural principles to achieve modularity, maintainability, and efficient data management.

Users interact with the system through the presentation layer, which consists of JavaFX interfaces developed using FXML and styled using CSS. User actions are processed by controller classes that implement the system's business logic and coordinate interactions among different modules.

Database operations are performed through DAO classes that communicate with the MySQL database using JDBC connectivity. Model classes represent the core business entities of the system, including customers, products, sales transactions, inventory records, warehouses, deliveries, and suppliers.

This integrated architecture ensures clear separation of responsibilities, facilitates future system enhancements, simplifies maintenance activities, and improves the overall scalability and reliability of the application.

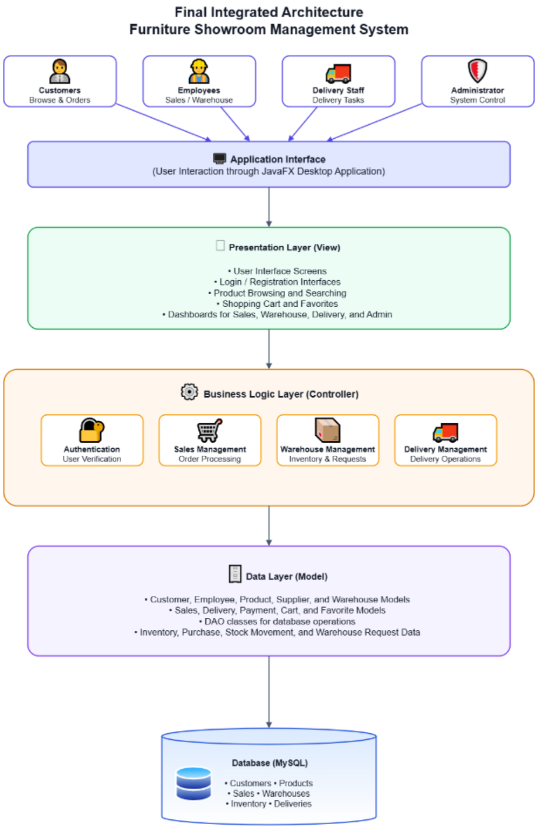


Figure4 : Final Integrated Architecture of the Furniture Showroom Management System

• Chapter Four: System Modeling

4.1 Business Workflow

The business workflow describes the high-level process followed within the showroom, starting from customer order placement and ending with product delivery. The workflow involves coordination between customers, sales employees, warehouse managers, and delivery employees. Warehouse verification is performed before order approval to ensure product availability, after which payment and delivery processes are completed. Inventory and system records are automatically updated throughout the process to maintain data consistency.

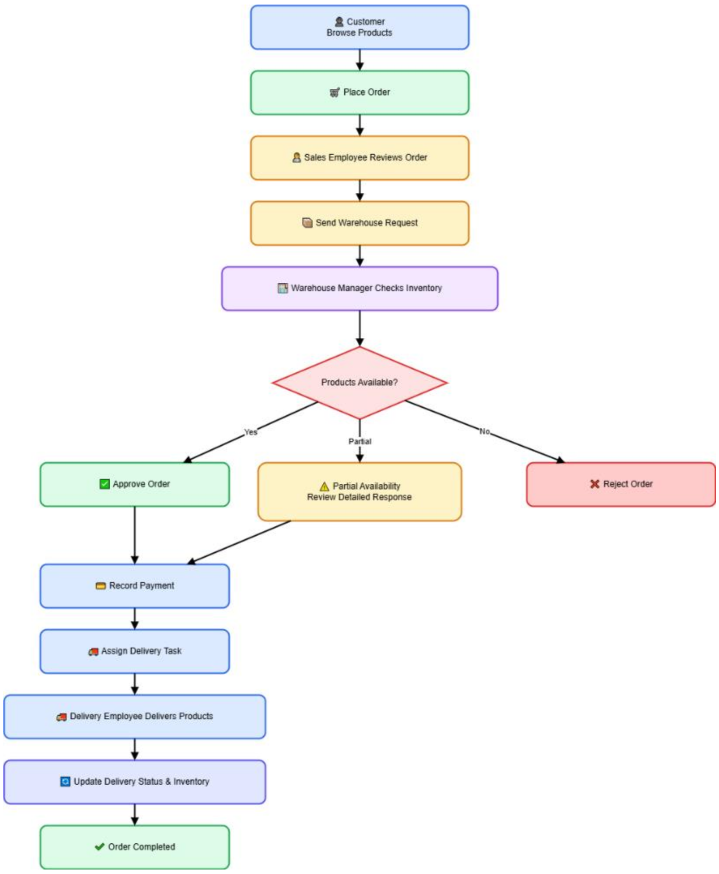


Figure 5 :Business Workflow of the Furniture Showroom Management System

4.2 Expected Entities and Relationships

After analyzing the business requirements of the Furniture Showroom Management System, the main entities of the system were identified based on the real operations performed inside a furniture showroom. These entities represent the core business objects involved in customer management, product management, sales processing, warehouse operations, supplier management, delivery tracking, and system authentication.

The purpose of this section is to provide a conceptual overview of the expected entities and their major relationships before presenting the detailed ER and EER diagrams. Detailed database constraints, weak entities, associative entities, participation constraints, and relational schema representation are discussed in later sections of this chapter.

4.3 Activity Diagram

The Activity Diagram illustrates the workflow of the Furniture Showroom Management System by showing the sequence of activities performed by customers, employees, and the system during major business operations such as order processing, warehouse verification, inventory updates, and delivery management.

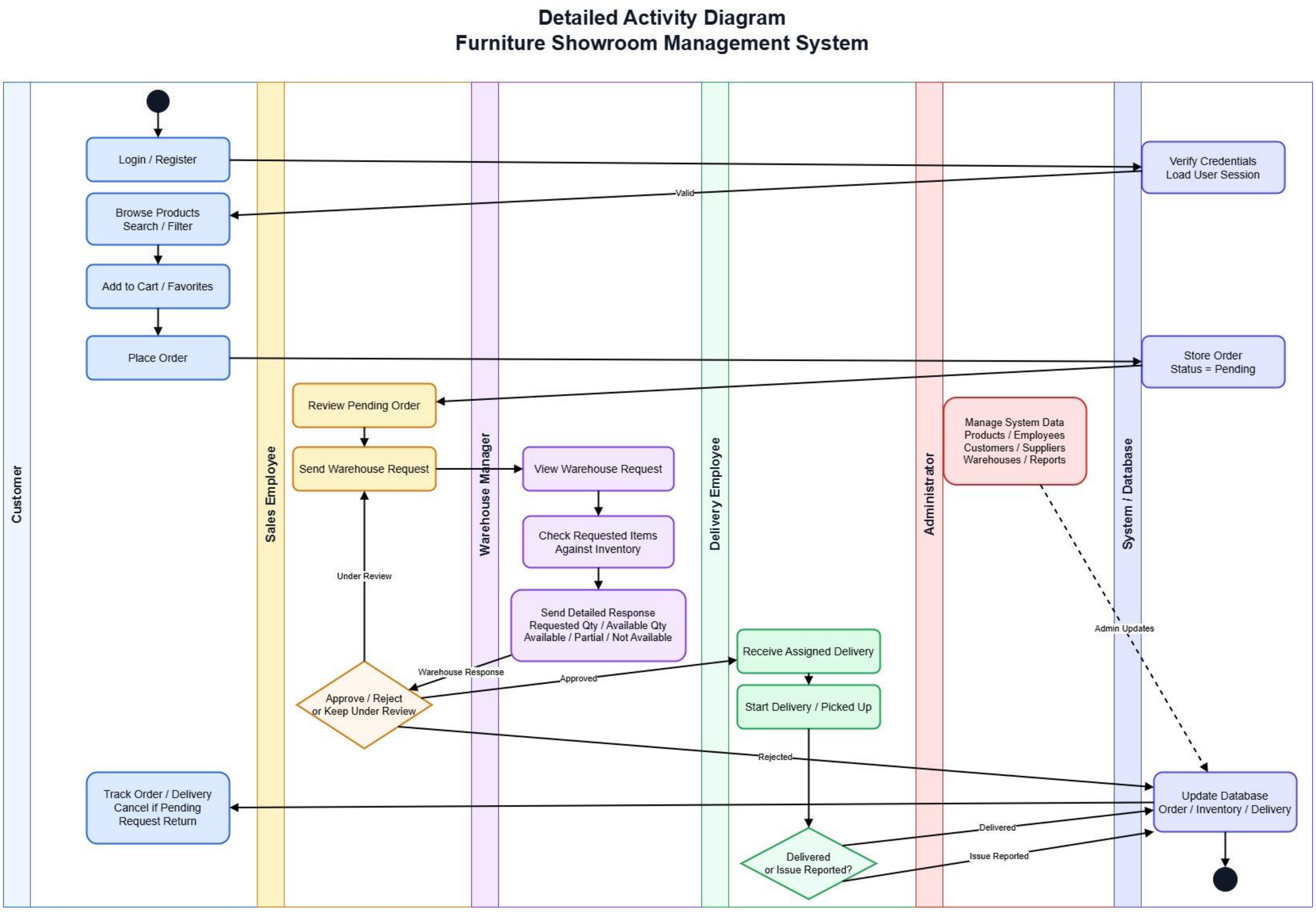


Figure 6: presents the detailed activity diagram of the Furniture Showroom Management System.

4.4 Use Case Diagram

The Use Case Diagram provides a high-level representation of the functional requirements of the Furniture Showroom Management System from the users' perspective. It identifies the main actors interacting with the system and illustrates the services and functionalities available to each actor.

The system consists of five primary actors: Customer, Sales Employee, Warehouse Manager, Delivery Employee, and Administrator. Each actor performs a specific set of operations according to their responsibilities within the furniture showroom environment.

Customers interact with the system to browse products, manage shopping carts and favorite lists, place orders, track order status, cancel pending orders, and request product returns. Sales employees are responsible for reviewing customer orders, communicating with warehouse managers, processing warehouse responses, and monitoring sales activities. Warehouse managers verify product availability, manage inventory records, and respond to warehouse requests. Delivery employees manage delivery operations and update delivery statuses, while administrators supervise the overall system by managing products, employees, customers, suppliers, and reports.

The use case diagram clearly defines the system boundaries and illustrates the interactions between different actors and the services provided by the Furniture Showroom Management System.

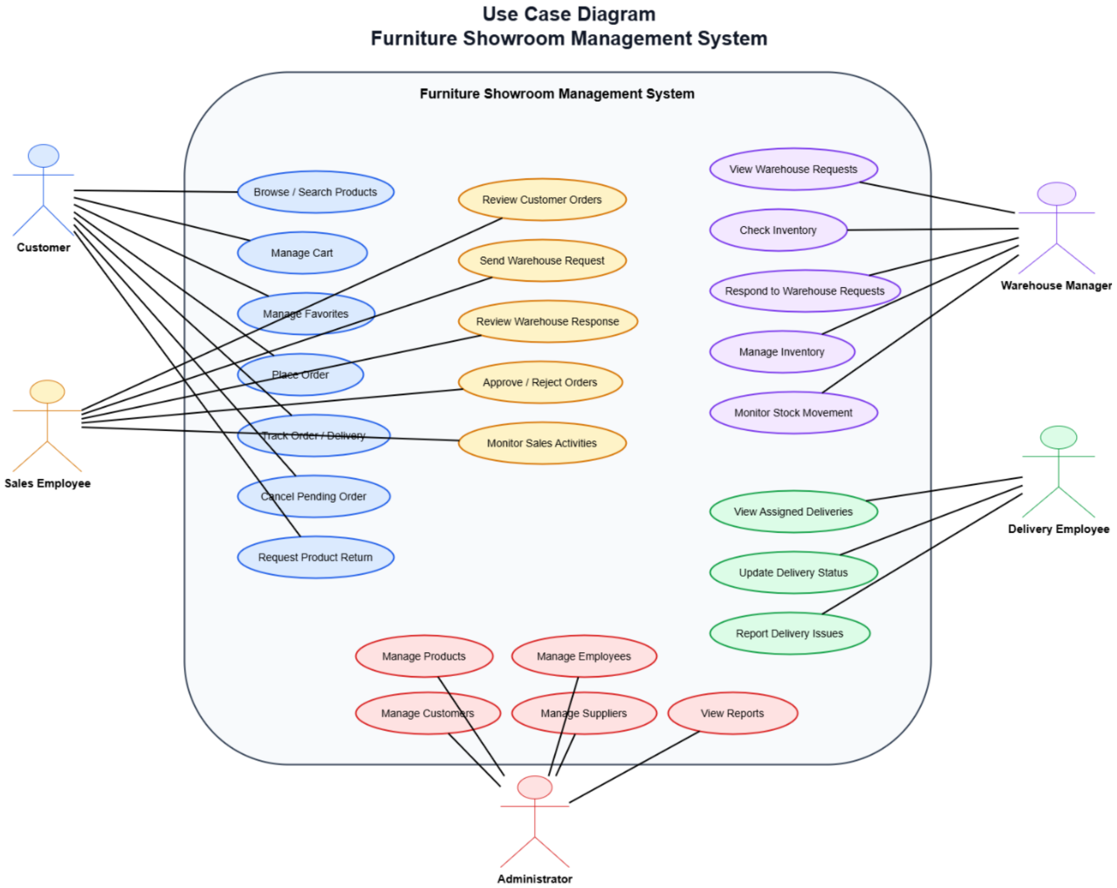


Figure 7:presents the Use Case Diagram of the Furniture Showroom Management System.

4.5 Sequence Diagram

The sequence diagram illustrates the chronological interaction between the system actors and internal components during the execution of a specific business process. It emphasizes the order in which messages are exchanged and demonstrates how different users and system modules cooperate to accomplish a particular task.

In the Furniture Showroom Management System, the customer order processing scenario was selected as the primary sequence because it represents the core business workflow of the showroom. This scenario involves multiple actors, including the customer, sales employee, warehouse manager, delivery employee, and the system database.

The sequence begins when a customer places an order through the system. The order is then reviewed by the sales employee, who sends a warehouse request to verify product availability. The warehouse manager checks the inventory and sends a detailed response indicating product availability, requested quantities, and available quantities. Based on the warehouse response, the sales employee either approves or rejects the order. Approved orders are assigned to delivery employees for shipment, and the delivery status is continuously updated in the system database.

The sequence diagram provides a detailed representation of message flow and interaction order among system participants during order processing.

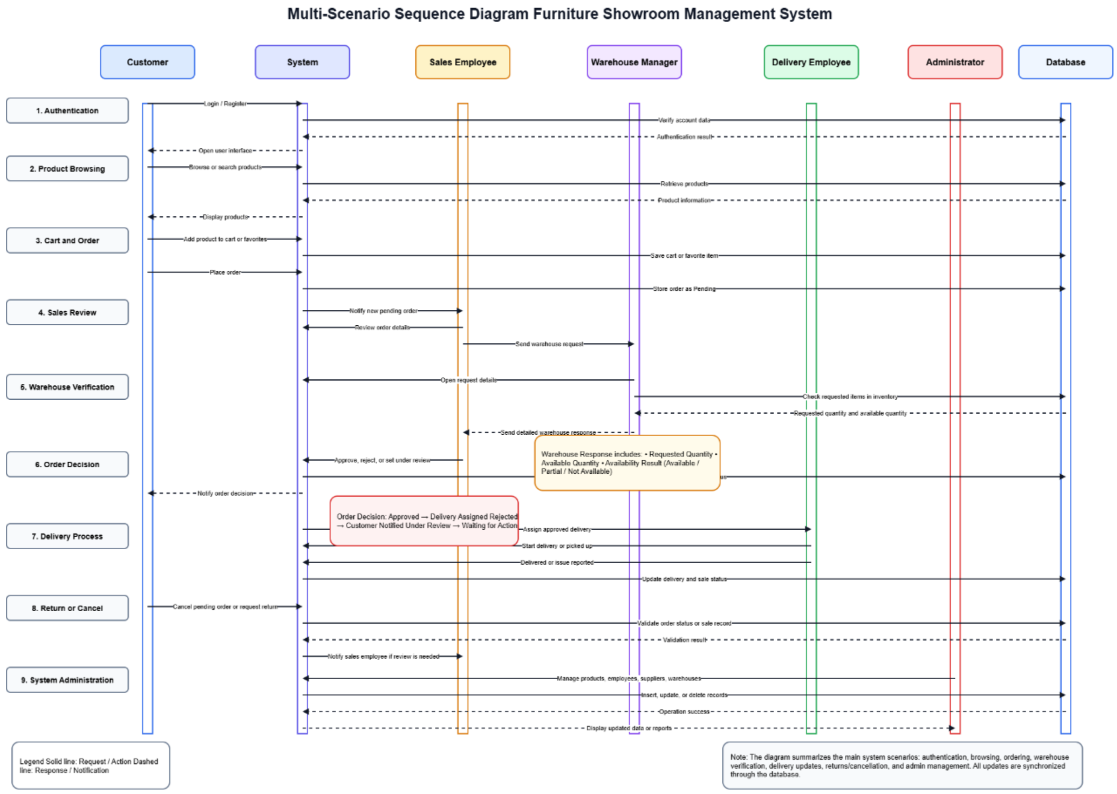


Figure 8: presents the Sequence Diagram for customer order processing in the Furniture Showroom Management System

4.6 ER Diagram

The Entity Relationship (ER) Diagram represents the conceptual database design of the Furniture Showroom Management System. It identifies the major entities, their attributes, primary keys, and the relationships among them.

The ER model was developed after analyzing the business requirements and operational workflow of the furniture showroom. It includes entities related to customer management, employee management, products, categories, suppliers, warehouses, inventory records, sales transactions, deliveries, payments, purchase transactions, and warehouse requests.

The diagram illustrates the cardinality constraints between entities, including one-to-one (1:1), one-to-many (1:M), and many-to-many (M:N) relationships. Associative entities were introduced to resolve many-to-many relationships and ensure proper database normalization.

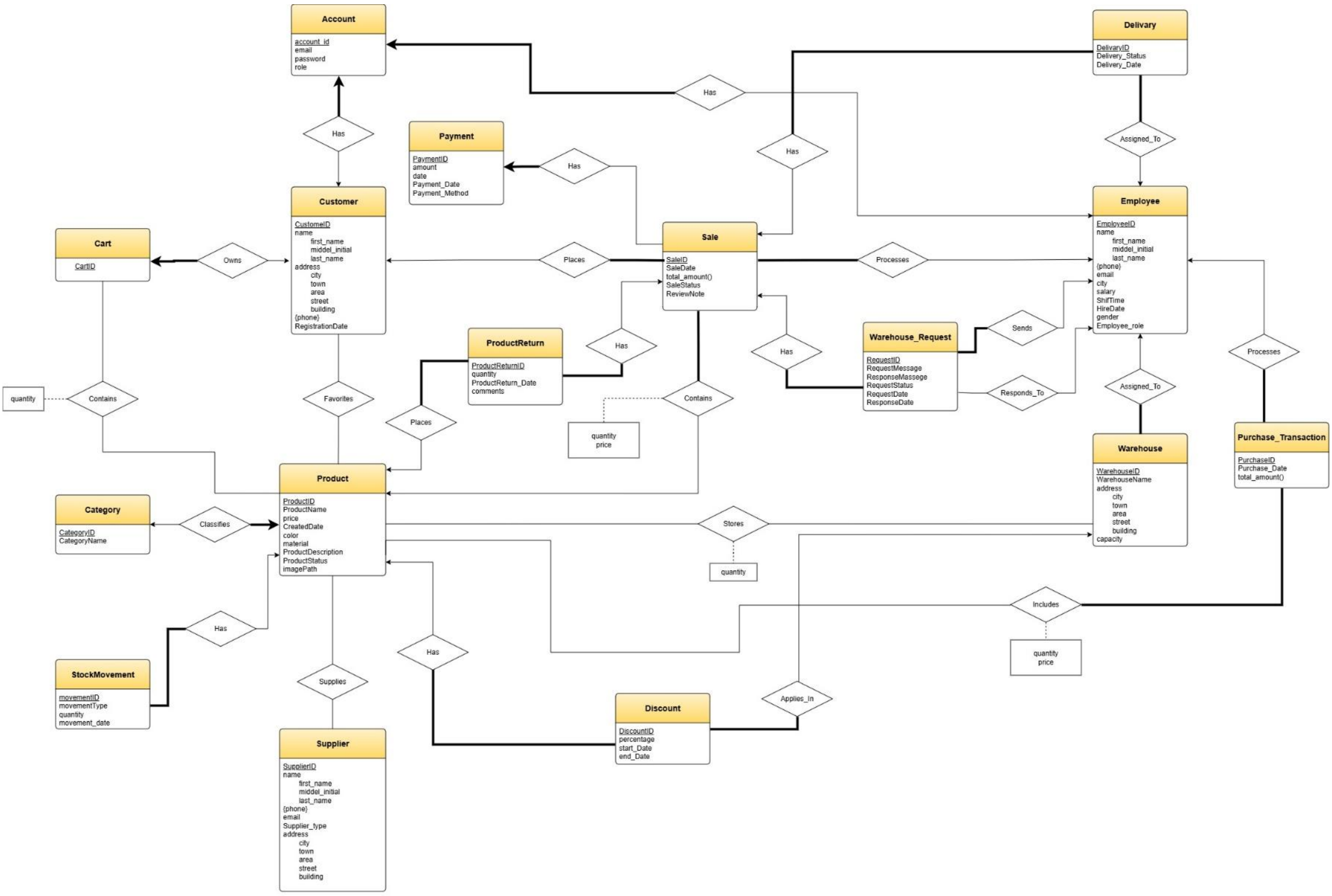


Figure 9 : presents the ER Diagram of the Furniture Showroom Management System

ER Design Notes

- Associative entities were introduced to resolve many-to-many relationships.
- Multi-valued attributes were modeled using separate entities.
- Primary and foreign keys were identified according to business requirements.

4.7 Relational Schema

The relational schema represents the logical database structure derived from the ER and EER models. Each entity is transformed into a relation (table), while relationships are represented using primary keys, foreign keys, and associative tables when necessary.

The Furniture Showroom Management System contains relations for customer management, employee management, authentication, product management, sales processing, warehouse operations, inventory management, supplier management, purchasing, delivery management, and payment processing.

Special attention was given to resolving many-to-many relationships through associative relations such as CartItem, Favorite, SaleDetails, Product_Supplier, Purchase_Details, and Inventory. These relations ensure data integrity while maintaining efficient data storage and retrieval.

The relational schema provides a direct blueprint for implementing the database using MySQL and supports all business operations performed by the Furniture Showroom Management System.

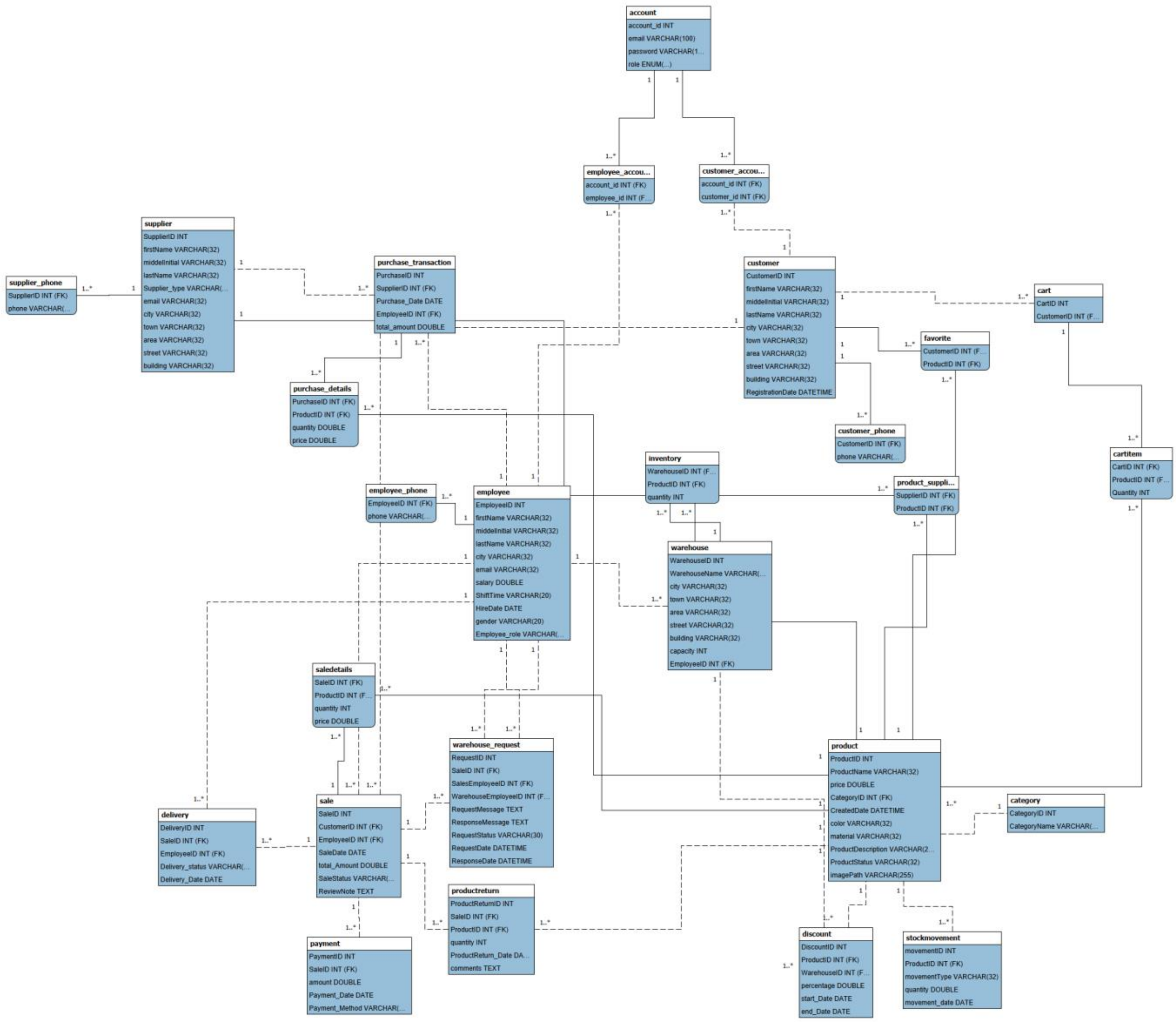


Figure 10:presents the Relational Schema of the Furniture Showroom Management System.

4.8 Normalization

Database normalization was applied to the Furniture Showroom Management System database to reduce data redundancy, eliminate update anomalies, and improve data consistency. The normalization process was performed according to the principles of relational database design and functional dependency analysis. The final database schema was verified to satisfy Third Normal Form (3NF), ensuring that every non-key attribute depends only on the primary key and not on other non-key attributes.

4.8.1 Functional Dependencies

The following functional dependencies were identified in the database schema:

- **Customer**

CustomerID → FirstName, MiddleInitial, LastName,
City, Town, Area, Street, Building, RegistrationDate

- **Employee**

EmployeeID → FirstName, MiddleInitial, LastName,
City, Email, Salary, ShiftTime, HireDate,
Gender, Employee_Role

- **Product**

ProductID → ProductName, Price, CategoryID,
CreatedDate, Color, Material,
ProductDescription, ProductStatus, ImagePath

- **Supplier**

SupplierID → FirstName, MiddleInitial, LastName,
Supplier_Type, Email, City, Town,
Area, Street, Building

- **Warehouse**

WarehouseID → WarehouseName, City, Town,
Area, Street, Building, Capacity, EmployeeID

- **Sale**

SaleID → CustomerID, EmployeeID,
SaleDate, Total_Amount,
SaleStatus, ReviewNote

These dependencies were used to verify that all relations satisfy the required normalization rules.

4.8.2 First Normal Form (1NF)

A relation is in First Normal Form (1NF) if all attributes contain atomic values and there are no repeating groups or multi-valued attributes. The purpose of 1NF is to ensure that each field stores only a single value and that every record can be uniquely identified.

During the design of the Furniture Showroom Management System, special attention was given to attributes that may contain multiple values. Phone numbers represent a common example of a multi-valued attribute because a customer, employee, or supplier may have more than one phone number.

A non-normalized design could store multiple phone numbers within the same relation as follows:

Customer(CustomerID, Name, Phone1, Phone2, Phone3)

Such a design creates repeating groups and violates the requirements of First Normal Form because the number of phone attributes is not fixed and multiple values are stored for the same type of information.

To eliminate this problem, phone numbers were separated into independent relations:

Customer_Phone(CustomerID, Phone)

Employee_Phone(EmployeeID, Phone)

Supplier_Phone(SupplierID, Phone)

Each phone number is stored in a separate tuple and linked to its corresponding entity through a foreign key relationship. This design allows any customer, employee, or supplier to have one or more phone numbers without introducing redundancy or violating normalization rules.

As a result, all attributes in the database contain atomic values, repeating groups were removed, and the database successfully satisfies the requirements of First Normal Form (1NF).

4.8.3 Second Normal Form (2NF)

A relation is considered to be in Second Normal Form (2NF) if it is already in First Normal Form (1NF) and every non-key attribute is fully functionally dependent on the entire primary key. This rule is particularly important for relations that use composite primary keys.

In the Furniture Showroom Management System, special attention was given to associative relations such as SaleDetails, Purchase_Details, Inventory, Product_Supplier, Favorite, and CartItem. These relations use composite keys and were analyzed to ensure that no non-key attribute depends on only part of the composite key.

Inventory is a representative example. A non-normalized design could store warehouse and product descriptive information directly within the Inventory relation, as shown below:

Inventory(WarehouseID, ProductID, WarehouseName, ProductName, Quantity)

In this design, WarehouseName depends only on WarehouseID, while ProductName depends only on ProductID. Since both attributes depend on only part of the composite key rather than the entire key, partial dependencies would exist, violating Second Normal Form.

To eliminate these partial dependencies, warehouse information was stored in the Warehouse relation and product information was stored in the Product relation. The Inventory relation was therefore reduced to:

Inventory(WarehouseID, ProductID, Quantity)

As a result, the remaining non-key attribute (Quantity) depends on the complete composite key (WarehouseID, ProductID), satisfying the requirements of Second Normal Form.

The same principle was applied to SaleDetails and Purchase_Details. Product descriptive information such as ProductName was maintained in the Product relation rather than being duplicated within transactional relations. Consequently, all non-key attributes in these relations depend on the entire composite key, ensuring compliance with 2NF.

4.8.4 Third Normal Form (3NF)

A relation is in Third Normal Form (3NF) if it is already in Second Normal Form (2NF) and contains no transitive dependencies. In other words, every non-key attribute must depend directly on the primary key and not on another non-key attribute.

During the database design process, several entities were separated to eliminate transitive dependencies and reduce data redundancy. This approach ensures that descriptive information is stored only once and can be maintained consistently throughout the system.

A representative example is the relationship between Product and Category. A non-normalized design could store category information directly within the Product relation:

Product(ProductID, ProductName, Price, CategoryID, CategoryName)

In this design, CategoryName depends on CategoryID rather than directly on ProductID. Therefore, the following transitive dependency exists:

ProductID → CategoryID
CategoryID → CategoryName

This creates a transitive dependency because CategoryName is determined indirectly through CategoryID.

To eliminate this dependency, category information was separated into an independent relation:

Category(CategoryID, CategoryName)

Product(ProductID, ProductName, Price, CategoryID, CreatedDate, Color, Material, ProductDescription, ProductStatus, ImagePath)

As a result, Product stores only the foreign key CategoryID, while all category details are maintained in the Category relation.

A similar design principle was applied throughout the database. Employee information is stored separately from Warehouse, customer information is stored separately from Sale transactions, and authentication data is isolated within the Account relation rather than being duplicated across Customer and Employee entities.

This separation prevents unnecessary redundancy and ensures that changes to one entity do not require updates in multiple relations.

Consequently, all non-key attributes in the Furniture Showroom Management System depend only on their respective primary keys. Transitive dependencies were removed, redundancy was minimized, and the final database schema satisfies the requirements of Third Normal Form (3NF).

4.8.5 Boyce-Codd Normal Form (BCNF)

Boyce-Codd Normal Form (BCNF) is a stronger version of Third Normal Form (3NF). A relation is considered to be in BCNF if, for every non-trivial functional dependency X → Y, the determinant X is a superkey of the relation.

After applying First, Second, and Third Normal Forms, the relations of the Furniture Showroom Management System were analyzed to verify compliance with BCNF. The analysis showed that all functional dependencies are determined by candidate keys or primary keys.

For example:

ProductID → ProductName, Price, Color, Material, CategoryID

CustomerID → Customer Information

EmployeeID → Employee Information

(SaleID, ProductID) → Quantity, Price

In each case, the determinant is either a primary key or a candidate key. Therefore, all relations satisfy the requirements of Boyce-Codd Normal Form (BCNF).

As a result, the database schema eliminates redundancy caused by functional dependencies and provides a robust and consistent relational design.

4.8.6 Normalization Result

The normalization process was successfully applied to the Furniture Showroom Management System database. The database design was systematically refined through multiple normalization stages to eliminate redundancy, improve consistency, and maintain data integrity.

First Normal Form (1NF) was achieved by removing repeating groups and multi-valued attributes through the creation of dedicated relations for customer, employee, and supplier phone numbers. This ensured that all attributes contain atomic values.

Second Normal Form (2NF) was achieved by eliminating partial dependencies from relations with composite primary keys, such as SaleDetails, Purchase_Details, and Inventory. Descriptive information related to products, warehouses, and suppliers was stored in their respective entities rather than being duplicated within associative relations.

Third Normal Form (3NF) was achieved by removing transitive dependencies through the separation of descriptive entities such as Category, Employee, Warehouse, Customer, Supplier, and Account. Each non-key attribute depends directly on the primary key of its relation.

Finally, the database schema was verified against Boyce-Codd Normal Form (BCNF). The analysis confirmed that all functional dependencies are determined by candidate keys or primary keys, indicating that the schema satisfies BCNF requirements.

As a result, the final database design minimizes redundancy, prevents insertion, update, and deletion anomalies, improves maintainability, and provides a reliable foundation for managing customers, products, sales, inventory, warehouse operations, deliveries, payments, and supplier transactions within the Furniture Showroom Management System.

• Chapter Five: System Implementation

5.1 System Implementation Introduction

This chapter presents the implementation of the Furniture Showroom Management System based on the requirements and models discussed in the previous chapters. The system was developed as a desktop application using JavaFX for the graphical user interface and MySQL as the database management system. The implementation follows a layered architecture and applies the Data Access Object (DAO) pattern to separate the business logic from database operations.

The system provides an integrated environment for managing furniture showroom activities, including product management, customer management, sales processing, warehouse operations, delivery tracking, supplier management, and administrative functions. Each module was implemented according to the business requirements identified during the analysis phase and modeled through the system modeling and database design diagrams presented in Chapter Four.

The implementation focuses on providing a user-friendly interface, efficient database interaction, secure account management, and seamless communication between different system actors. Customers can browse products, manage shopping carts and favorites, place orders, track deliveries, and request returns. Employees are provided with specialized interfaces based on their responsibilities, including sales processing, warehouse inventory management, and delivery operations. Administrators can manage system resources and monitor overall business activities through reports and management tools.

The following sections describe the implementation details of the database, user interfaces, system modules, and the interaction mechanisms that enable the complete workflow of the Furniture Showroom Management System.

5.2 Database Implementation

5.2.1 Database Overview

The Furniture Showroom Management System uses MySQL as its relational database management system. The database was designed according to the relational schema developed during the database design phase and normalized to reduce redundancy and maintain data consistency.

The database stores all information related to customers, employees, products, suppliers, warehouses, inventory, sales transactions, deliveries, payments, discounts, warehouse requests, product returns, and user accounts. Foreign key constraints were used to preserve referential integrity between related tables and ensure the validity of stored data.

The database structure supports the complete operational workflow of the showroom, from product acquisition and inventory management to sales processing, delivery tracking, and customer service operations.

5.2.2 Main Database Tables

The Furniture Showroom Management System is built around a set of core business tables that represent the primary entities involved in showroom operations. These tables store the essential data required to manage customers, employees, products, sales transactions, warehouse operations, suppliers, and financial activities:

Table 2: Main Database Tables

Table Name	Description
Customer	Stores customer personal information and registration details.
Employee	Stores employee information, roles, salaries, and work details.
Account	Stores login credentials and account roles used for authentication.
Product	Stores furniture product information including pricing, material, color, images, and availability status.
Category	Stores product categories used for classification and filtering.
Supplier	Stores supplier information and contact details.
Warehouse	Stores warehouse information, locations, capacities, and assigned managers.
Sale	Stores customer orders and sales transaction information.
Delivery	Stores delivery assignments and delivery status information.
Payment	Stores payment records associated with customer orders.
Purchase_Transaction	Stores purchasing transactions made from suppliers.
ProductReturn	Stores product return requests submitted by customers.
Warehouse_Request	Stores communication and verification requests between sales employees and warehouse managers before order approval.
Discount	Stores product discount information and promotional campaigns.
StockMovement	Stores inventory movement records, including stock additions and stock withdrawals.

These tables form the foundation of the database and support the major business processes implemented within the system.

5.2.3 Associative and Supporting Tables

Several many-to-many relationships exist within the system. To properly implement these relationships and maintain database normalization, associative tables were introduced. These tables connect related entities while storing additional relationship-specific information when required.

Table 3: Associative and Supporting Tables

Table Name	Purpose
SaleDetails	Resolves the many-to-many relationship between Sale and Product while storing quantity and selling price information.
Product_Supplier	Resolves the many-to-many relationship between Supplier and Product.
Purchase_Details	Resolves the many-to-many relationship between Purchase_Transaction and Product while storing purchased quantities and prices.
Inventory	Resolves the many-to-many relationship between Warehouse and Product while storing available quantities in each warehouse.
Favorite	Resolves the many-to-many relationship between Customer and Product and stores customer favorite products.
CartItem	Resolves the many-to-many relationship between Cart and Product while storing product quantities selected by customers.
Product_Supplier	Resolves the many-to-many relationship between Supplier and Product.

The use of associative tables reduces redundancy and allows the database to efficiently represent complex business relationships.

5.2.4 Supporting Tables

In addition to the core and associative tables, several supporting tables were implemented to manage multivalued attributes, account mappings, and additional system functionality.

Table 4: Supporting Tables

Table Name	Description
Customer_Phone	Stores multiple phone numbers for customers.
Employee_Phone	Stores multiple phone numbers for employees.
Supplier_Phone	Stores multiple phone numbers for suppliers.
Customer_Account	Links customer records to authentication accounts.
Employee_Account	Links employee records to authentication accounts.

Cart	Stores shopping carts associated with customers before order confirmation.

These supporting tables enhance the flexibility of the database design and contribute to maintaining a normalized and scalable database structure.

5.2.5 Database Integrity Constraints

To ensure data accuracy, consistency, and reliability, several integrity constraints were implemented within the Furniture Showroom Management System database. These constraints help maintain valid relationships between entities and prevent incorrect or inconsistent data from being stored.

Primary Key Constraints

Each table contains a primary key that uniquely identifies every record within the table. Examples include CustomerID, EmployeeID, ProductID, SaleID, WarehouseID, and SupplierID.

Foreign Key Constraints

Foreign keys were used to establish relationships between related tables and maintain referential integrity. For example, the Sale table references both Customer and Employee tables, while the Inventory table references Warehouse and Product tables.

NOT NULL Constraints

Mandatory attributes were defined using NOT NULL constraints to ensure that essential information is always provided. Examples include customer names, product names, sale dates, warehouse capacities, and payment amounts.

UNIQUE Constraints

Unique constraints were implemented where duplicate values are not allowed. For example, account email addresses must be unique to prevent multiple accounts from using the same email address.

CHECK Constraints

Several business rules were enforced using CHECK constraints. Examples include employee roles, customer account roles, product status values, sale status values, delivery status values, supplier types, and warehouse request statuses.

Cascading Operations

ON DELETE CASCADE was applied in selected relationships to automatically remove dependent records when the parent record is deleted. This prevents orphan records and maintains database consistency.

These constraints collectively improve data quality, enforce business rules, and support the reliable operation of the system.

5.2.6 Database Relationships

The database contains several relationship types that represent the business rules of the Furniture Showroom Management System. These relationships were implemented using primary keys, foreign keys, and associative tables. The following table summarizes the main database relationships according to their cardinality.

Table 5: Database Relationships and Relationship Resolution

Relationship Type	Relationship	Implementation / Resolution
One-to-One (1:1)	Customer — Cart	Cart.CustomerID is UNIQUE and references Customer.CustomerID.
One-to-One (1:1)	Customer — Account	Resolved using Customer_Account(account_id, customer_id), where each customer is linked to one account.
One-to-One (1:1)	Employee — Account	Resolved using Employee_Account(account_id, employee_id), where each employee is linked to one account.
One-to-One (1:1)	Sale — Delivery	Delivery.SaleID references Sale.SaleID.
One-to-Many (1:M)	Customer — Customer_Phone	Customer_Phone.CustomerID references Customer.CustomerID.
One-to-Many (1:M)	Employee — Employee_Phone	Employee_Phone.EmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Supplier — Supplier_Phone	Supplier_Phone.SupplierID references Supplier.SupplierID.
One-to-Many (1:M)	Category — Product	Product.CategoryID references Category.CategoryID.
One-to-Many (1:M)	Customer — Sale	Sale.CustomerID references Customer.CustomerID.
One-to-Many (1:M)	Employee — Sale	Sale.EmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Sale — Payment	Payment.SaleID references Sale.SaleID.
One-to-Many (1:M)	Sale — ProductReturn	ProductReturn.SaleID references Sale.SaleID.
One-to-Many (1:M)	Product — ProductReturn	ProductReturn.ProductID references Product.ProductID.
One-to-Many (1:M)	Product — StockMovement	StockMovement.ProductID references Product.ProductID.
One-to-Many (1:M)	Product — Discount	Discount.ProductID references Product.ProductID.

One-to-Many (1:M)	Warehouse — Discount	Discount.WarehouseID references Warehouse.WarehouseID.
One-to-Many (1:M)	Supplier — Purchase_Transaction	Purchase_Transaction.SupplierID references Supplier.SupplierID.
One-to-Many (1:M)	Employee — Purchase_Transaction	Purchase_Transaction.EmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Warehouse Manager Employee — Warehouse	Warehouse.EmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Sale — Warehouse_Request	Warehouse_Request.SaleID references Sale.SaleID.
One-to-Many (1:M)	Sales Employee — Warehouse_Request	Warehouse_Request.SalesEmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Warehouse Employee — Warehouse_Request	Warehouse_Request.WarehouseEmployeeID references Employee.EmployeeID.
One-to-Many (1:M)	Delivery Employee — Delivery	Delivery.EmployeeID references Employee.EmployeeID.
Many-to-Many (M:N)	Cart — Product	Resolved using CartItem(CartID, ProductID, Quantity).
Many-to-Many (M:N)	Customer — Product	Resolved using Favorite(CustomerID, ProductID).
Many-to-Many (M:N)	Sale — Product	Resolved using SaleDetails(SaleID, ProductID, quantity, price).
Many-to-Many (M:N)	Supplier — Product	Resolved using Product_Supplier(SupplierID, ProductID).
Many-to-Many (M:N)	Purchase_Transaction — Product	Resolved using Purchase_Details(PurchaseID, ProductID, quantity, price).
Many-to-Many (M:N)	Warehouse — Product	Resolved using Inventory(WarehouseID, ProductID, quantity).

This relationship structure ensures referential integrity, supports normalized database design, and accurately represents the real operational relationships within the showroom.

5.2.7 Database Relationships

The database design incorporates advanced EER concepts including specialization, generalization, participation constraints, and associative entities to accurately represent the business rules of the Furniture Showroom Management System.

- **Employee Specialization**

The Employee entity acts as a superclass and is specialized into the following employee roles:

- Sales Employee
- Warehouse Manager
- Delivery Employee

The specialization follows a **Disjoint** constraint because an employee can belong to only one employee role at a time.

The specialization also follows a **Total** constraint because every employee stored in the system must have a defined role.

- **Account Specialization**

System accounts are associated with either customers or employees through:

- Customer_Account
- Employee_Account

This design separates authentication information from business entities and supports role-based access control.

- **Participation Constraints**

Several relationships enforce total participation.

Examples include:

- Every Cart must belong to a Customer.
- Every CartItem must belong to a Cart.
- Every SaleDetails record must belong to a Sale.
- Every Inventory record must reference both a Product and a Warehouse.

Other entities such as Product, Customer, and Supplier may exist independently without participating in all relationships, resulting in partial participation in some cases.

- **Associative Entities**

The database resolves many-to-many relationships using associative tables including:

- SaleDetails

- CartItem
- Favorite
- Product_Supplier
- Purchase_Details
- Inventory

These tables improve normalization and eliminate redundancy while preserving relationship-specific information.

5.3 User Interface Implementation

5.3.1 System Landing Page

The System Landing Page serves as the public entry point to the Furniture Showroom Management System. It introduces visitors to the showroom through a modern and visually appealing interface that reflects the identity and services of the furniture business.

The page provides quick access to the system through the **Enter System** button, which redirects users to the main application interfaces. Additionally, users can access an introductory promotional video through the **Watch Video** feature, allowing visitors to learn more about the showroom, its products, and its services before entering the system.

The interface also includes branding elements, welcoming messages, and navigation controls designed to improve user experience and create a professional first impression.

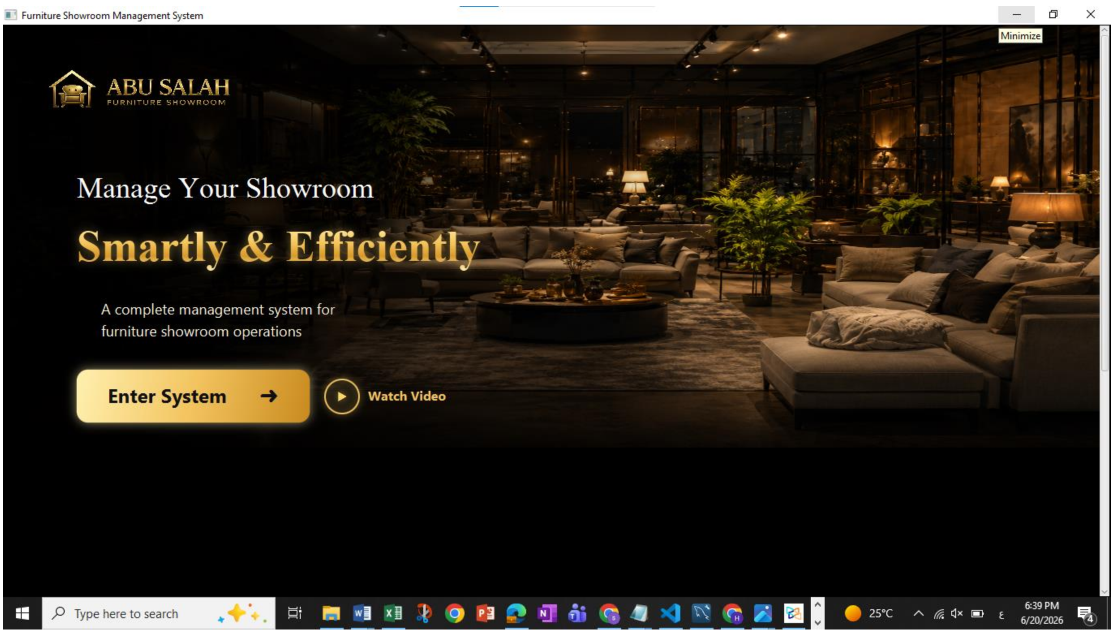


Figure 11:System Landing Page

5.3.2 Guest Product Browsing Interface

The Guest Product Browsing Interface allows visitors to explore available furniture products without requiring authentication. Guests can browse the product catalog, search for products, apply filters, view product prices, and access detailed product information.

The interface provides advanced filtering options based on category, material, color, price range, and product availability. Product cards display images, names, prices, and quick access to detailed product information.

Guests are allowed to freely explore the showroom inventory; however, purchasing-related actions require authentication. Features such as adding products to the shopping cart and managing favorite products are restricted to registered users. When a guest attempts to perform these actions, the system prompts the user to log in before continuing.

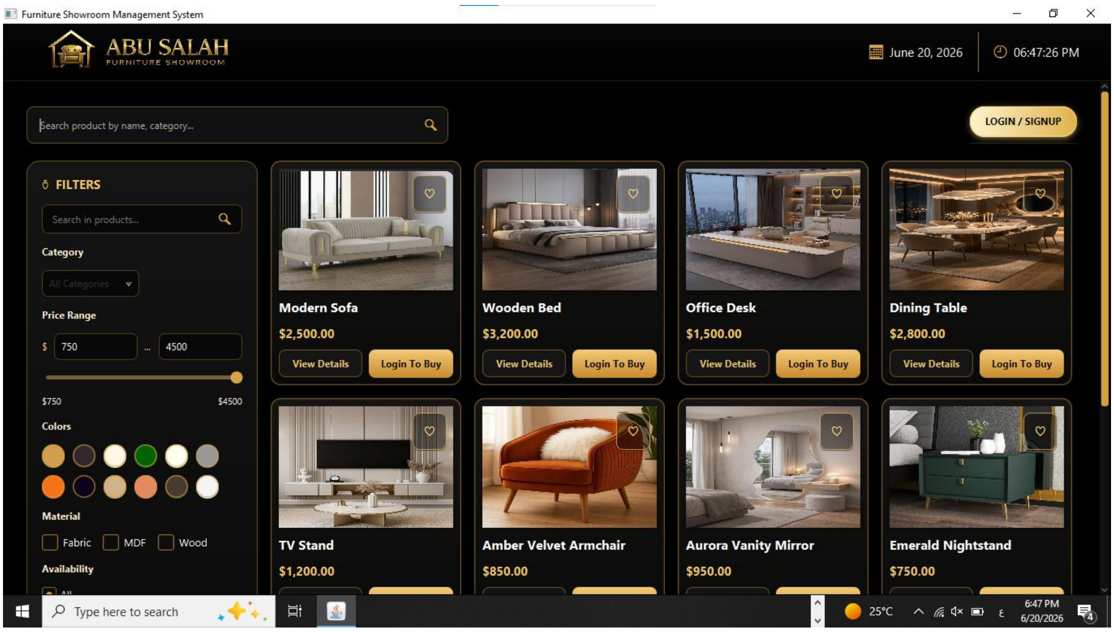


Figure 12: Guest Product Browsing Interface

5.3.3 Authentication and Account Management

The Authentication Interface provides secure access to the Furniture Showroom Management System for all registered users. Users can log in using their registered email address and password, and the system verifies the provided credentials against the stored account information in the database.

The authentication module supports multiple user roles, including customers, sales employees, warehouse managers, delivery employees, and administrators. After successful authentication, users are redirected to the appropriate interface according to their assigned role and permissions.

The interface also provides a **Continue as Guest** option, allowing visitors to access the guest browsing interface without creating an account. Guest users can explore products and view product information but cannot perform restricted operations such as purchasing products, managing favorites, or adding items to the shopping cart.

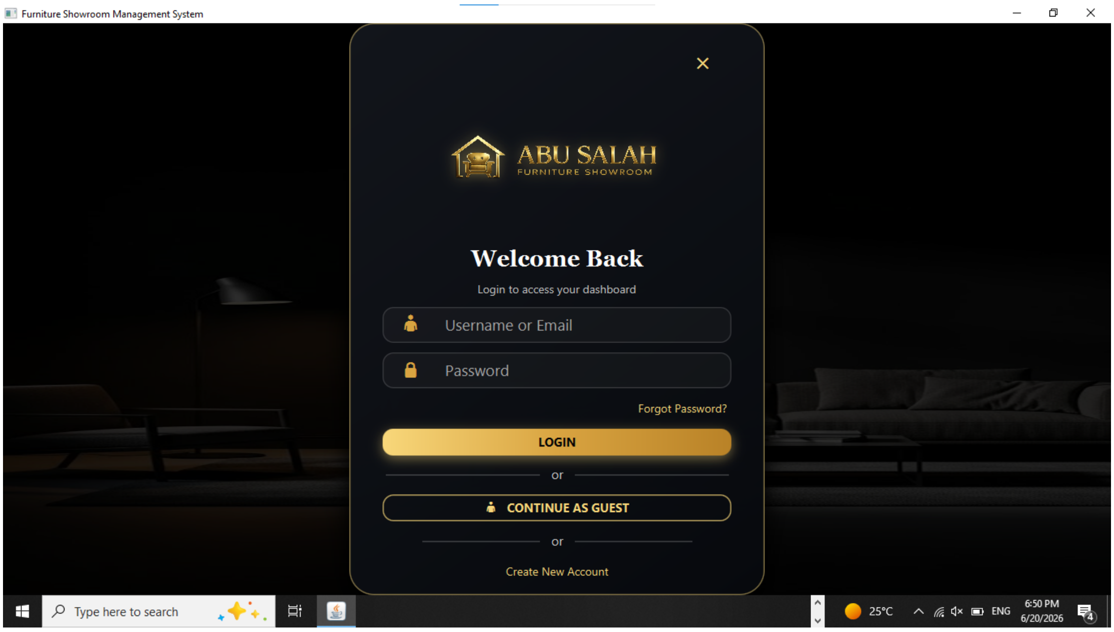


Figure 13: Authentication interface

5.3.4 Customer Registration Interface

The Customer Registration Interface allows new users to create customer accounts within the system. Users are required to provide personal information including their first name, last name, email address, phone number, and password.

The system validates all entered information before creating the account. Email addresses must be unique and cannot already exist in the database. Password confirmation is required to ensure accuracy during account creation.

After successful registration, the customer account is stored in the database and can be used for future authentication and purchasing operations.

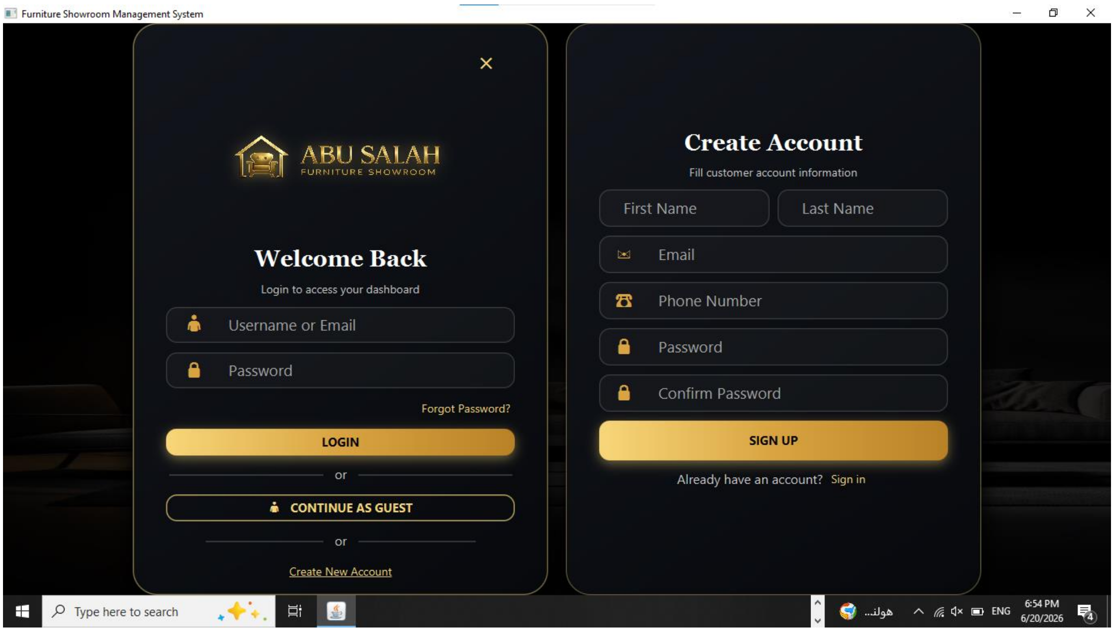


Figure 14: Customer Registration Interface

5.3.5 User Role Navigation

After successful authentication, the system automatically redirects users to the appropriate interface according to their assigned role. This role-based navigation ensures that each user can access only the functionalities relevant to their responsibilities.

The system currently supports five primary user roles:

Table 6:User Rule:

User Role	Redirected Interface
Customer	Customer Interface
Sales Employee	Sales Employee Dashboard
Warehouse Manager	Warehouse Manager Dashboard
Delivery Employee	Delivery Employee Dashboard
Administrator	Administrator Dashboard

Each dashboard provides specialized tools, permissions, and functionalities designed to support the operational responsibilities of the corresponding user role.

This role-based architecture enhances system security, improves usability, and prevents unauthorized access to restricted system functions.

5.3.5.1 Customer Interface

The Customer Interface provides registered customers with a complete shopping environment that allows them to browse furniture products, manage favorite items, maintain shopping carts, place orders, track deliveries, and submit product return requests.

After successful authentication, customers gain access to personalized features that are not available to guest users. These features improve the shopping experience by allowing customers to interact directly with products and monitor their purchases throughout the order lifecycle.

The customer module consists of several integrated interfaces, including product details, favorite products, shopping cart management, order tracking, and return request management.

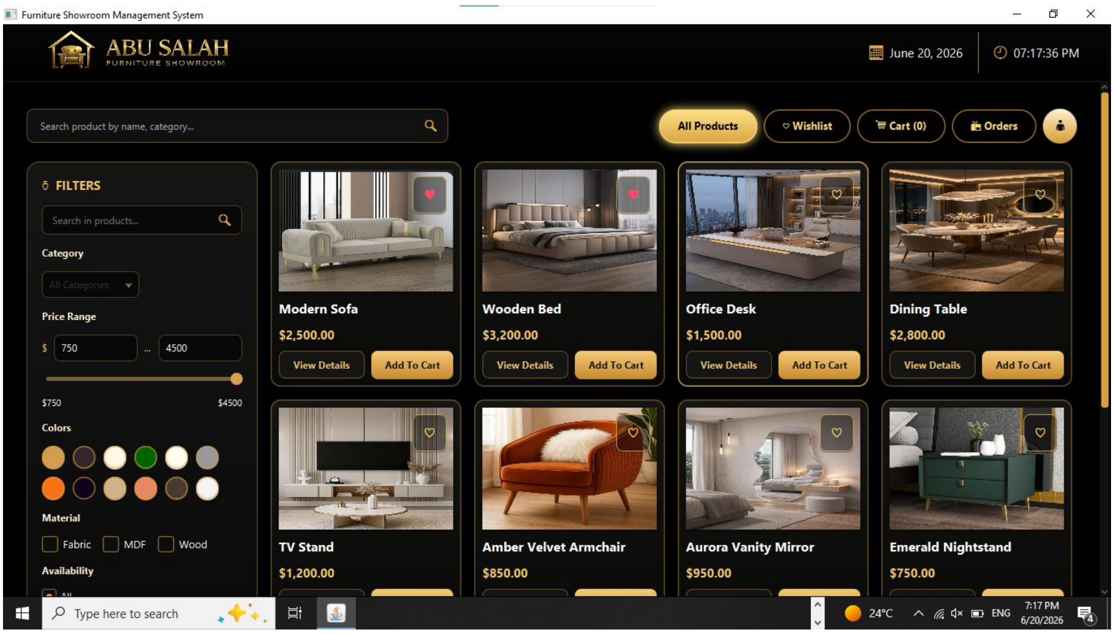


Figure 15: Customer Interface

5.3.5.1.1 Product Details Interface

The Product Details Interface presents complete information about a selected furniture product. Customers can view product specifications, price, category, material, color, description, availability status, and product image.

The interface allows authenticated customers to add products to their favorites list or shopping cart directly from the product page.

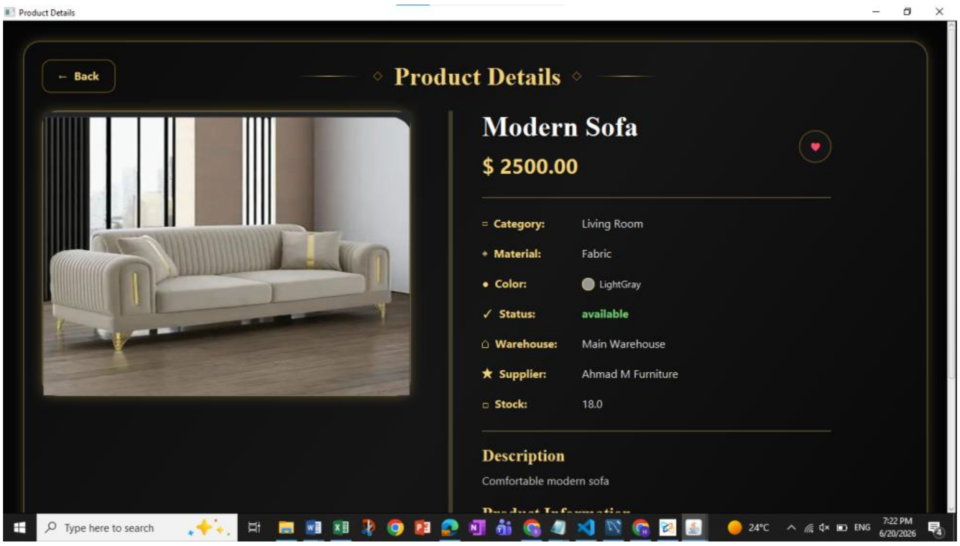


Figure 16:Product Details Interface

5.3.5.1.2 Favorite Products Interface

The Favorite Products Interface enables customers to maintain a personalized wishlist of preferred products. Customers can quickly access products they are interested in without searching for them repeatedly.

Products can be added or removed from the favorites list at any time, and changes are immediately reflected within the system.

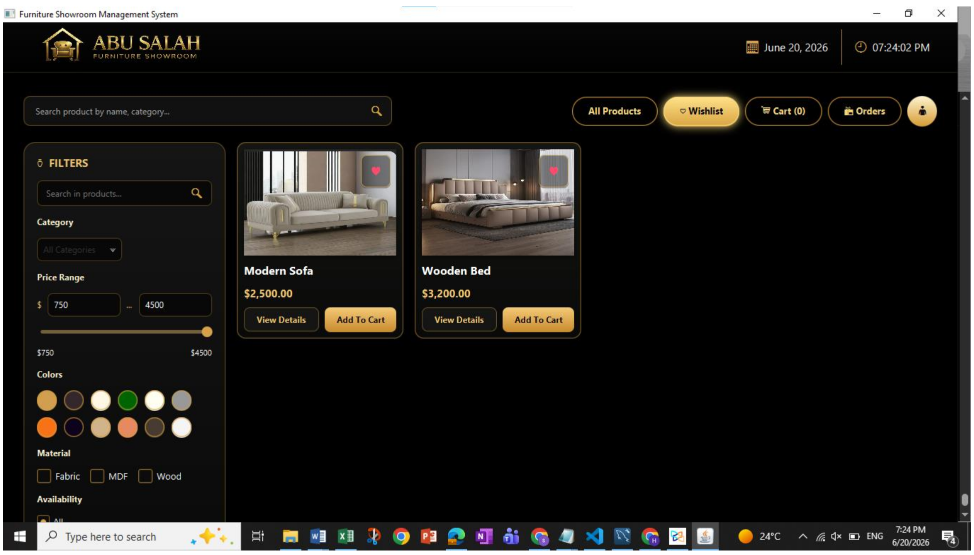


Figure 17: Favorite Products Interface

5.3.5.1.3 Shopping Cart Interface

The Shopping Cart and Checkout Interface enables customers to manage selected products before completing the purchase process. Customers can review product details, modify quantities, remove products from the cart, and monitor pricing information including product totals and overall order cost.

The interface automatically calculates the order subtotal, applied discounts, delivery fees, and final payable amount. Customers are also required to provide delivery information before proceeding with the checkout process.

The delivery section collects the customer's delivery address, including city, town, area, street, and building information. This information is used to generate delivery records and assign delivery tasks after order approval.

Once all information is verified, customers can proceed to checkout and submit the order for further processing by the sales employee and warehouse manager.

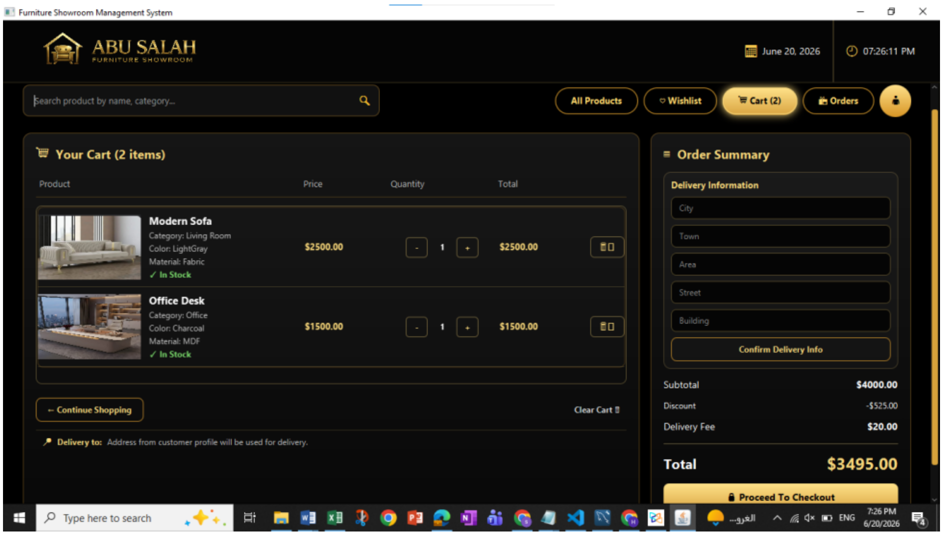


Figure 18: Shopping Cart and Checkout Interface

5.3.5.1.4 Order Review and Submission Interface

The Order Review and Submission Interface appears after the customer fills in the delivery information and proceeds from the shopping cart page. This interface allows the customer to review all order details before final submission.

At this stage, the order is still temporary and has not yet been confirmed or stored as a final order. The page displays customer information, contact details, delivery address, selected products, quantities, original prices, offered prices, savings, and the final offer summary.

When the customer clicks the **Submit Offer** button, the system confirms the order, clears the shopping cart, stores the order in the database, and adds it to the customer orders page with an initial status of **Pending**. The submitted order also becomes visible to the sales employee, who is responsible for reviewing and processing pending customer orders.

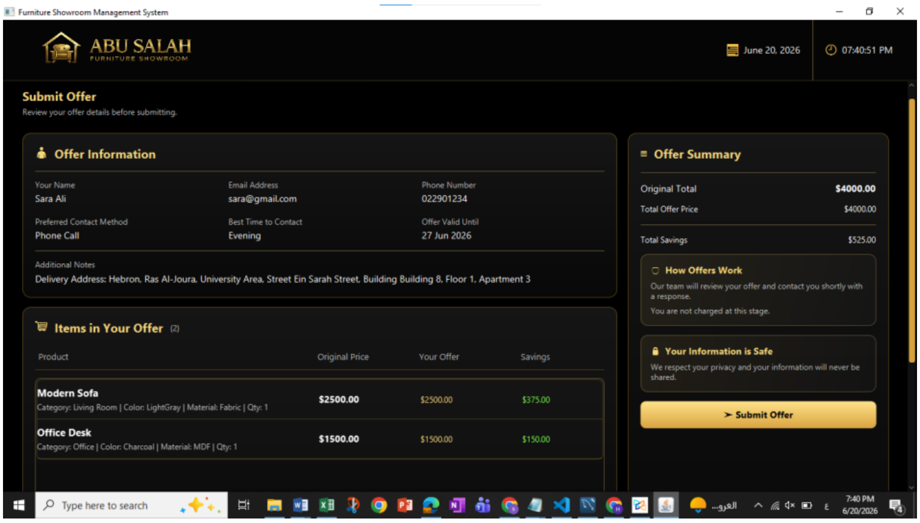


Figure 19: Order Review and Submission Interface

5.3.5.1.5 Customer Orders and Tracking Interface

The Customer Orders and Tracking Interface enables customers to monitor all submitted orders throughout their lifecycle. The interface provides a centralized view of order history, order statuses, delivery progress, and detailed order information.

Customers can search for specific orders, filter orders by status, define date ranges, and sort results according to their preferences. Summary cards provide quick statistics regarding the total number of orders as well as pending, delivered, and cancelled orders.

Selecting an order displays detailed information including order date, number of items, total amount, delivery status, purchased products, and order contents. Customers can track order progress from submission through warehouse verification and delivery completion.

Orders that are still under review may be cancelled directly by the customer before processing is completed.

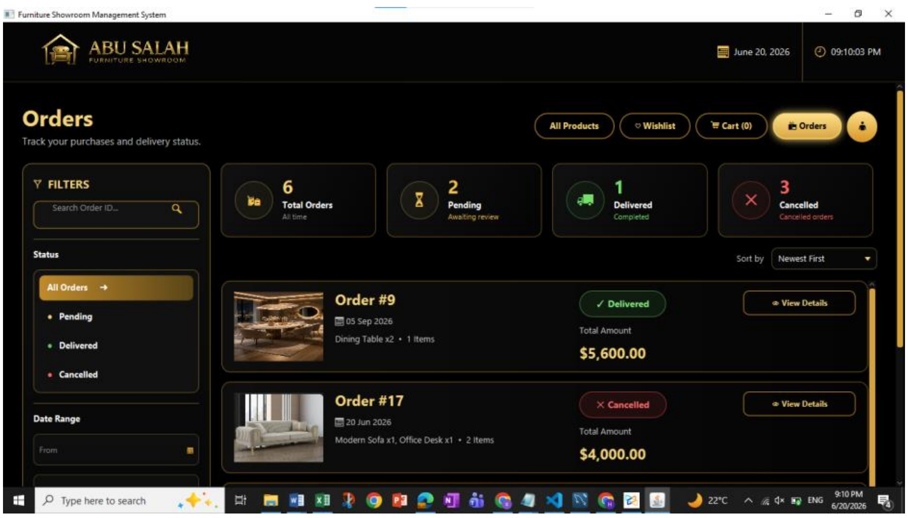


Figure 20: Customer Orders and Tracking Interface

5.3.5.2 Sales Employee Interface

The Sales Employee Interface enables sales staff to monitor customer orders, review warehouse responses, track sales activities, and manage order processing workflows. The interface provides real-time operational information and supports decision-making during order approval and fulfillment.

5.3.5.2.1 Sales Dashboard

The Sales Dashboard serves as the main workspace for sales employees. It provides a high-level overview of current sales activities, pending customer orders, payment status, warehouse-related notifications, and operational performance indicators.

The dashboard displays key performance metrics such as pending orders, approved orders, revenue, pending payments, and orders currently in progress. Priority alerts help sales employees identify urgent situations requiring immediate attention.

Employees can review alert details by selecting the corresponding alert action button. The system then displays detailed information about the selected alert category. To begin processing customer orders, the employee can use the **Start Reviewing Orders** action, which opens the Sales Workbench for detailed order management.

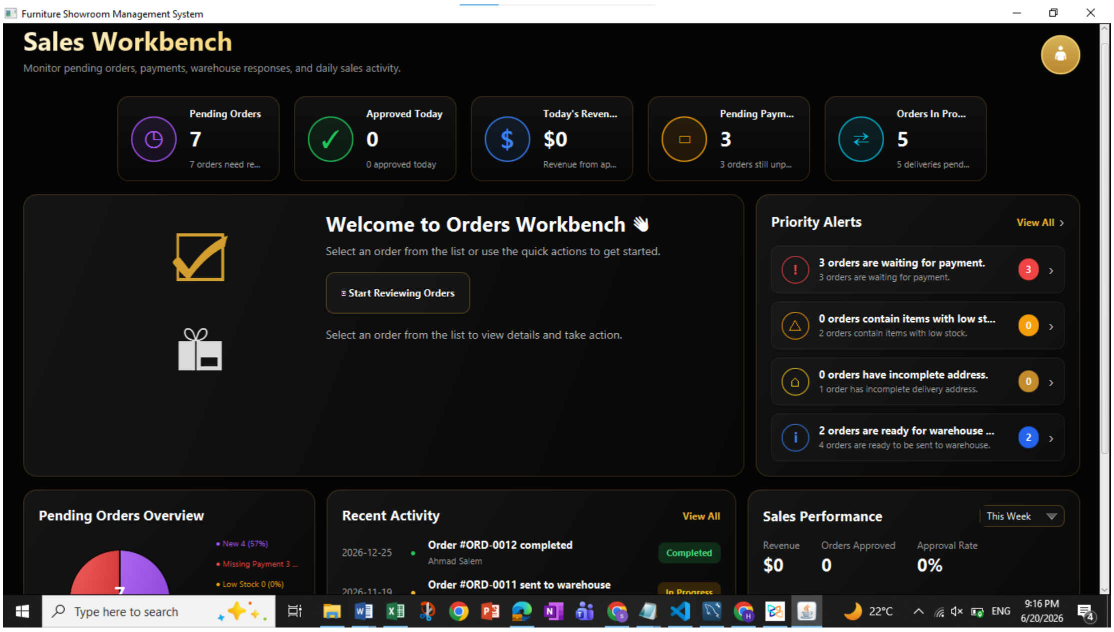


Figure 21: Sales Employee Dashboard

5.3.5.2.2 Sales Workbench Interface

The Sales Workbench Interface is the primary operational workspace used by sales employees to review, validate, and process customer orders. The interface provides complete visibility into customer information, order contents, payment status, warehouse responses, delivery information, and order history.

Orders are organized into three categories: Pending, Approved, and Rejected. Sales employees can search for specific orders, review order details, and perform the necessary actions required to move an order through the fulfillment process.

When a pending order is selected, the system displays detailed information including customer information, ordered products, payment records, validation results, warehouse responses, and delivery information.

The interface also provides a validation checklist that helps employees verify order completeness before making decisions regarding approval, rejection, or warehouse verification.

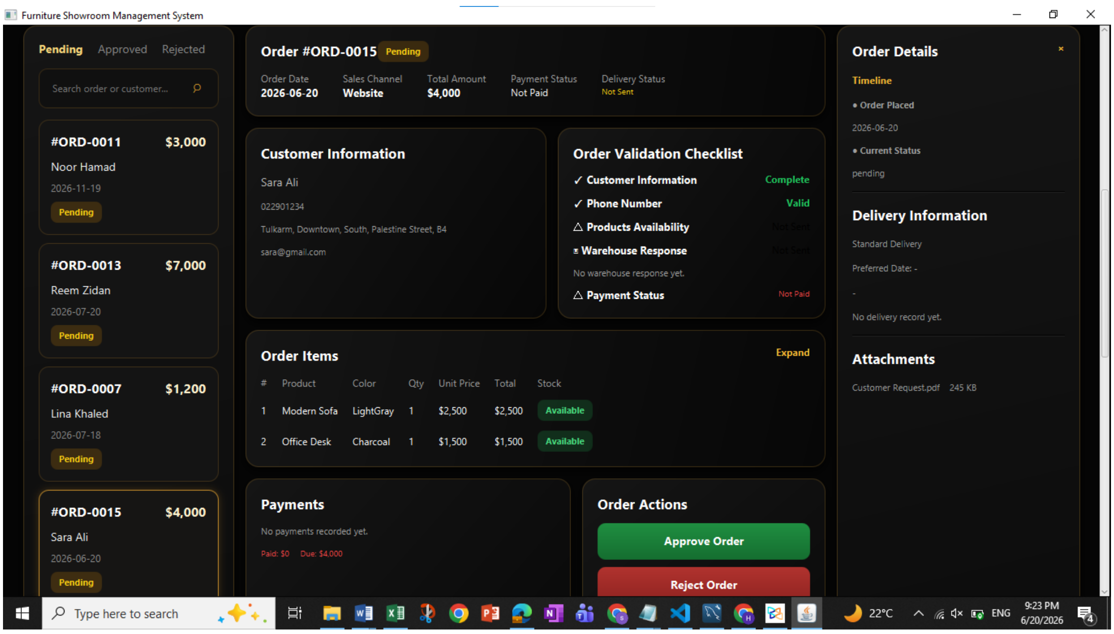


Figure 22: Sales Workbench Interface

Order Processing Actions

Sales employees can perform several actions during order processing:

Approve Order

Approves the customer order and updates the order status within the system. Approved orders proceed to the next stage of fulfillment and delivery processing.

Reject Order

Rejects the customer order and records the rejection decision within the system. Rejected orders become visible in the rejected orders section.

Send to Warehouse

Creates a warehouse request and forwards the order to the warehouse manager for inventory verification. The warehouse manager reviews requested products and returns a detailed response indicating requested quantities, available quantities, and availability results for each product.

Add Payment

Records customer payment information and updates payment status associated with the order.

Add Order Note

Allows sales employees to attach operational notes, comments, or review information related to the selected order.

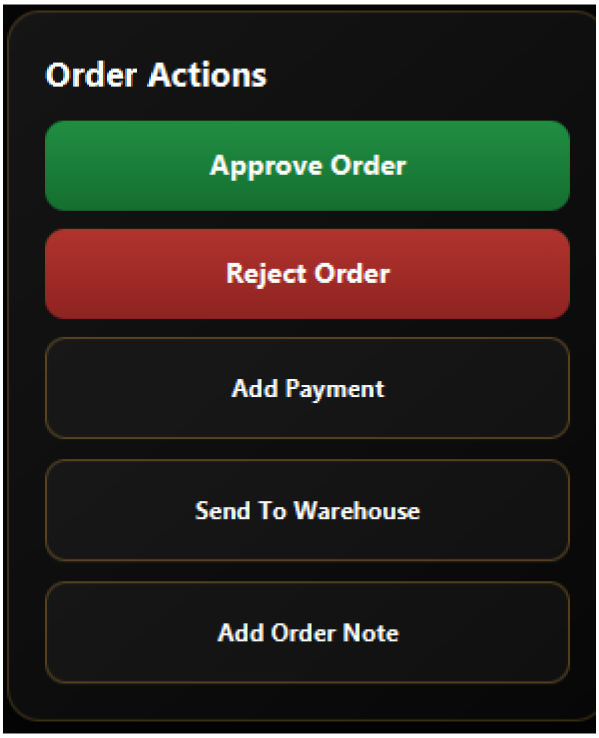


Figure 23:Order Processing Actions

Order Validation Checklist

The system automatically evaluates several validation criteria before an order is approved:

- Customer information completeness.
- Phone number validation.
- Product availability verification.
- Warehouse response availability.

- Payment status verification.

These checks assist sales employees in making informed decisions and reduce processing errors during order fulfillment.

5.3.5.3 Warehouse Manager Interface

5.3.5.3.1 Warehouse Manager Dashboard and Request Review

The Warehouse Manager Workbench serves as the central workspace for inventory verification and warehouse request processing. This interface enables the warehouse manager to monitor stock availability, review incoming requests from sales employees, and respond with detailed availability information before sales decisions are finalized.

The dashboard provides an overview of warehouse activities through key performance indicators, pending warehouse requests, inventory status charts, low-stock alerts, and stock movement statistics. Warehouse requests submitted by sales employees are displayed in a dedicated section where each request can be reviewed individually.

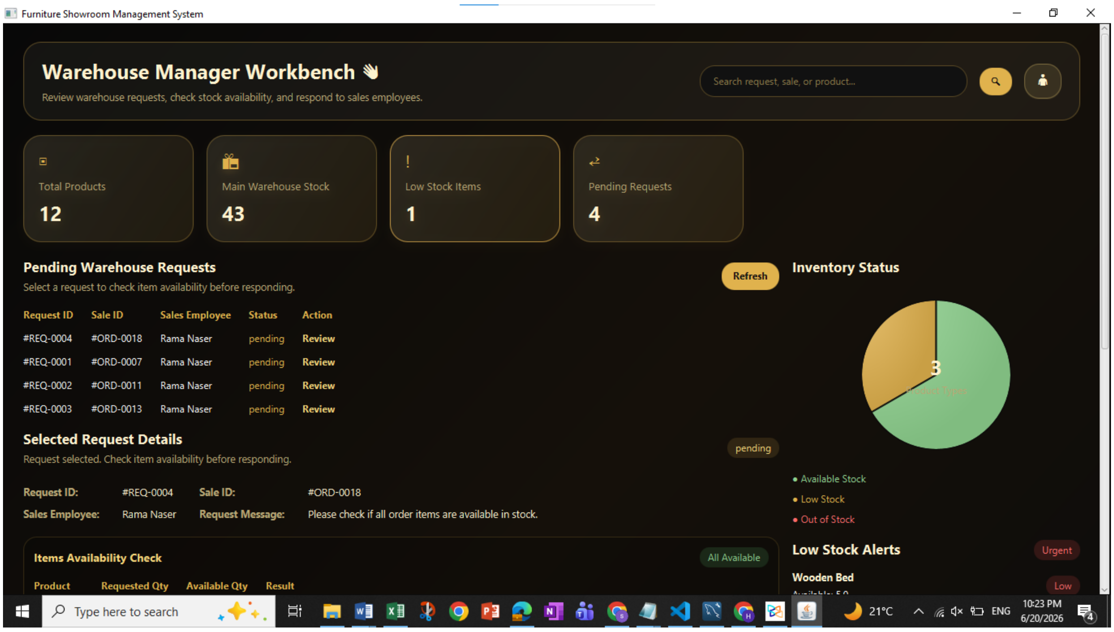


Figure 24: Warehouse Manager Workbench Dashboard

5.3.5.3.2 Warehouse Manager Dashboard and Request Review

When a warehouse request is selected, the system displays detailed request information including the request ID, associated sale ID, sales employee information, and the warehouse request message. The warehouse manager can then verify product availability by comparing requested quantities against available inventory quantities.

The system provides a detailed item availability table showing each requested product, requested quantity, available quantity, and the resulting availability status. Based on this analysis, the warehouse manager sends an appropriate response back to the sales employee.

Figure 25: Warehouse Request Processing Interface

5.3.5.4 Delivery Employee Interface

5.3.5.4.1 Delivery Workbench Dashboard

The Delivery Workbench serves as the main workspace for delivery employees. It provides access to assigned deliveries, customer information, delivery routes, and delivery status updates. The interface enables delivery employees to manage the complete delivery process from warehouse pickup until final delivery confirmation.

The dashboard displays delivery statistics, assigned orders, customer addresses, route information, delivery items, and recent delivery activities. It also provides quick actions that allow the delivery employee to update the status of assigned deliveries in real time.

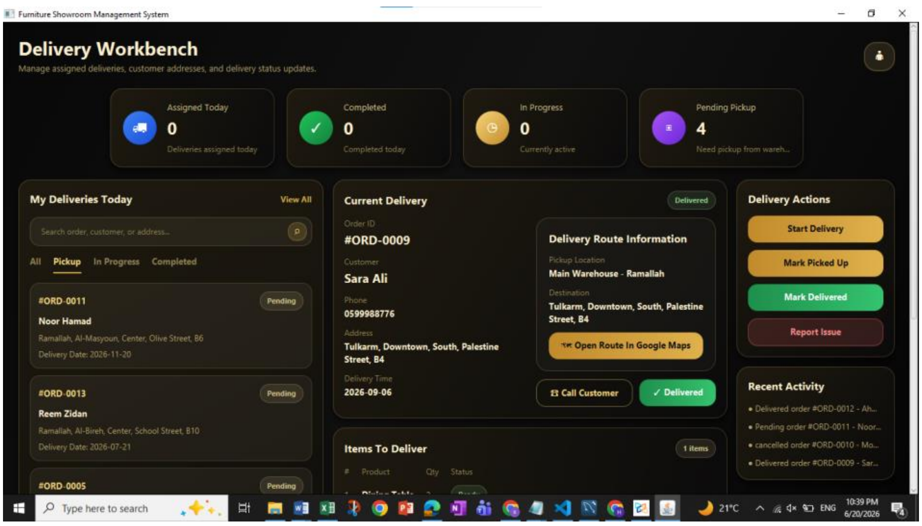


Figure 26:Delivery Employee Workbench

5.3.5.4.2 Delivery Status Management

The delivery employee is responsible for updating delivery progress throughout the delivery lifecycle. The system provides dedicated actions that allow the delivery employee to record the current delivery status and report any issues encountered during transportation.

The status updates are synchronized with the system database, allowing customers and other employees to monitor delivery progress in real time.

Available Actions

- **Start Delivery**
  - Changes the delivery status from Pending Pickup to In Progress.
- **Mark Picked Up**
  - Confirms that the products have been collected from the warehouse.
- **Mark Delivered**
  - Confirms successful delivery to the customer and updates the order status accordingly.
- **Report Issue**
  - Allows the delivery employee to report delivery problems such as customer unavailability, address issues, damaged products, or transportation delays.

Delivery Workflow

The delivery process follows the sequence below:

1. Delivery is assigned by the Sales Employee.
2. Delivery Employee receives the assigned order.
3. Products are collected from the warehouse.
4. Delivery status is updated to In Progress.
5. Products are transported to the customer location.
6. Delivery Employee marks the order as Delivered or reports an issue if delivery cannot be completed.

5.3.5.5 Administrator Interface

5.3.5.5.1 Administrator Dashboard

The Administrator Dashboard provides a comprehensive overview of the entire Furniture Showroom Management System. It allows administrators to monitor business performance, inventory levels, sales activities, customer statistics, warehouse operations, and employee-related information from a centralized interface.

The dashboard presents real-time business indicators, sales analytics, inventory alerts, delivery activities, and top-selling products, enabling management to make informed operational decisions.

Figure 27: Administrator Dashboard

5.3.5.5.2 Administrative Management Modules

The Administrator can access all management modules through the navigation menu. These modules provide full control over products, customers, employees, suppliers, warehouses, categories, purchases, and sales records. Since all management modules follow a similar CRUD (Create, Read, Update, Delete) structure, one representative example is presented below.

Example

The Product Management module allows administrators to manage showroom products. Administrators can add new products, update product information, delete products, search products, apply filters, and monitor product availability.

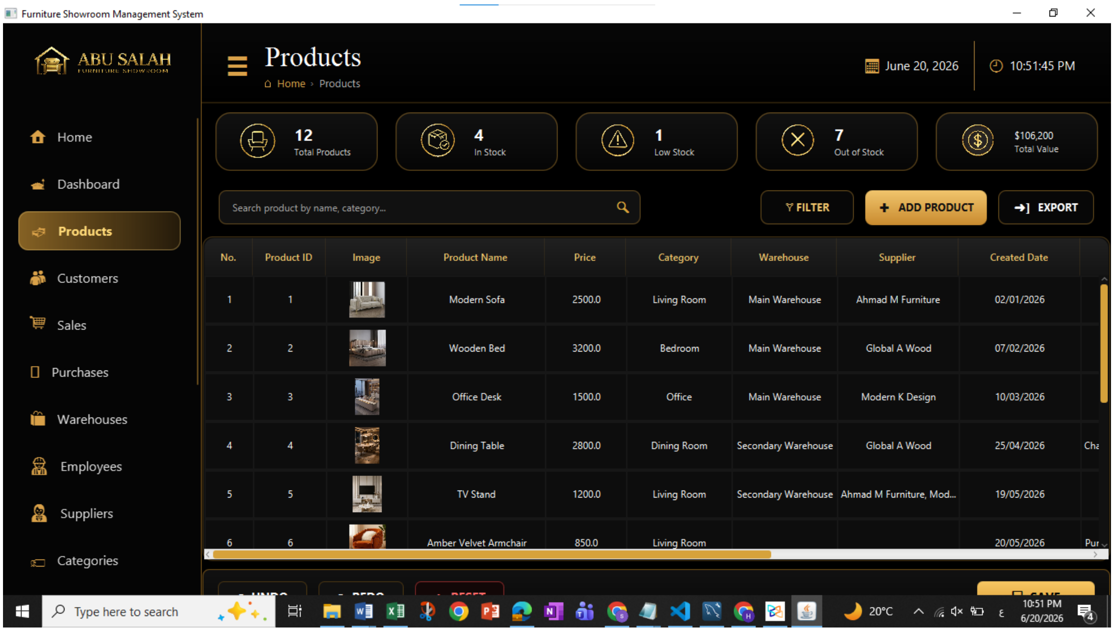


Figure 28:Product Management module

• Chapter Six: Testing

6.1 Test Scenario 1: Successful Order Approval and Delivery

Objective

To verify the complete order lifecycle, starting from customer order placement, followed by sales employee review and approval, and ending with successful delivery completion by the delivery employee.

- **Customer Order Submission:**

The customer successfully submitted a new order. The order appears in the customer orders page with a Pending Review status awaiting sales employee verification.

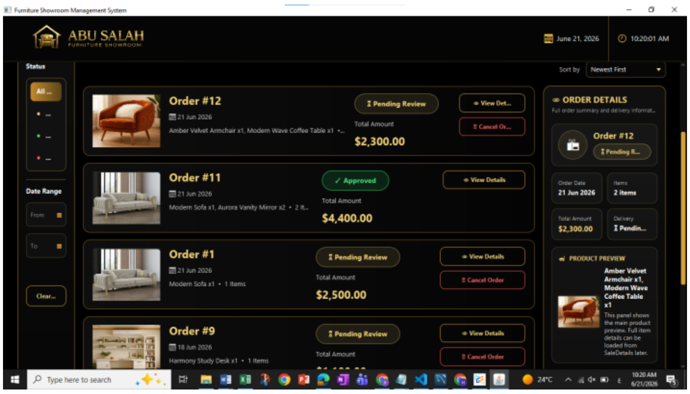


Figure 29:Customer Order Submission and Pending Review Status

- **Sales Employee Order Review**

Description

The order appears in the Sales Employee Workbench. Customer information, delivery information, order items, and validation checklist are displayed before approval.

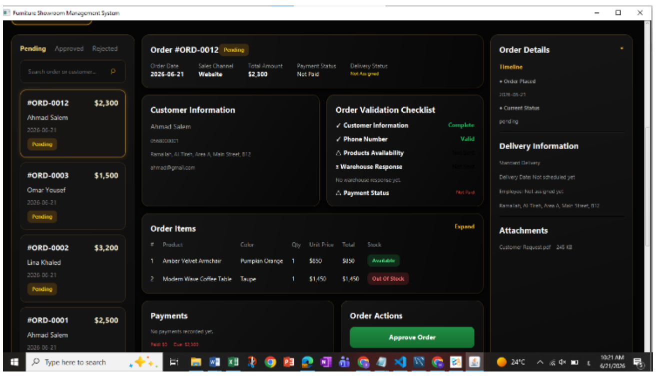


Figure 30: Sales Employee Order Review and Validation

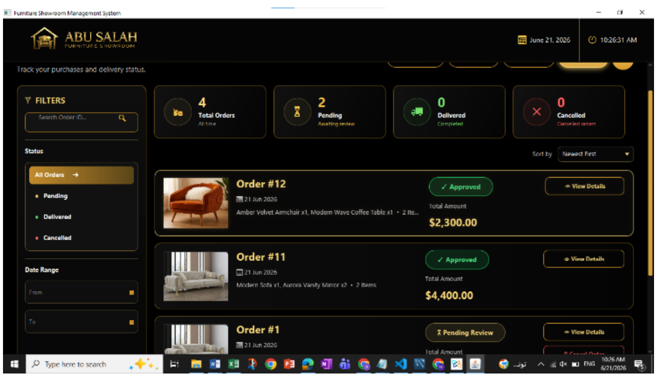


Figure 31: Order Approval Confirmation

- **Delivery Assignment**

Description

After approval, the system automatically creates a delivery assignment and displays the order in the Delivery Employee Workbench. Route information and customer details are available for delivery planning.

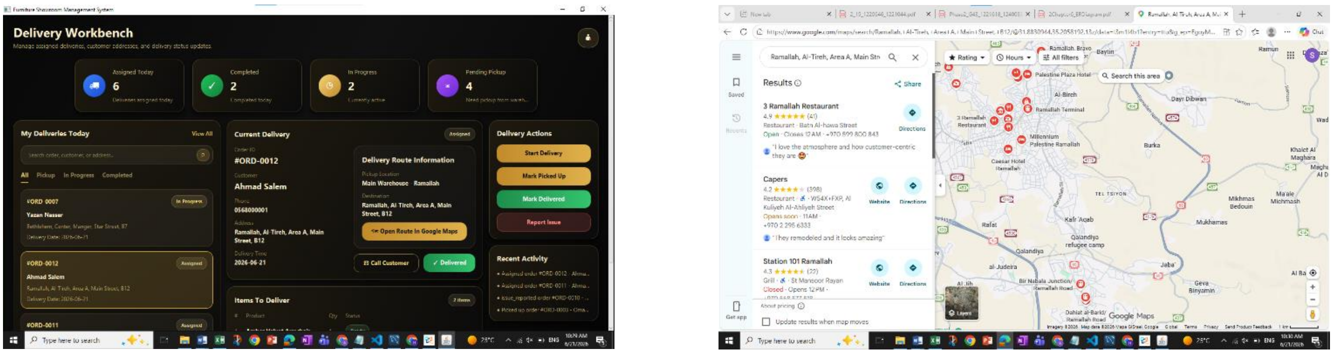


Figure 32:Delivery Assignment and Route Information

- **Delivery Status Updates:**

Description

The delivery employee successfully updated the delivery status throughout the delivery lifecycle. Database verification confirmed that each status transition was stored correctly.

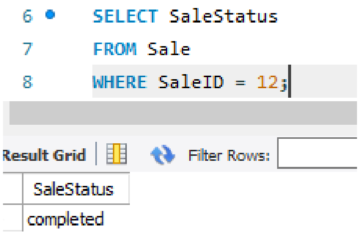


Figure 36: Delivery Status Updated to Completed

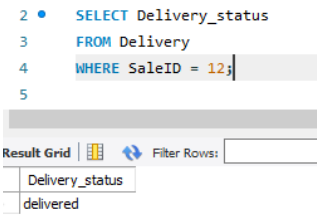


Figure 33:Delivery Status Updated to Delivered

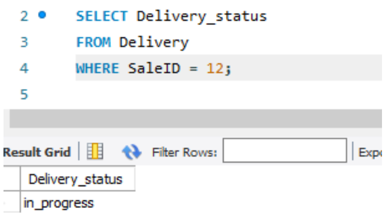


Figure 35:Delivery Status Updated to In Progress

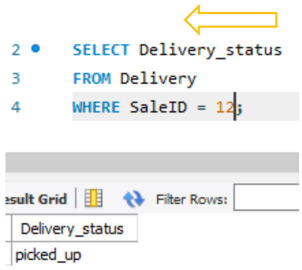


Figure 34:Delivery Status Updated to Picked Up

- **Customer Delivery Confirmation**

Description

After delivery completion, the customer can track the order and view the final Delivered status from the Orders page.

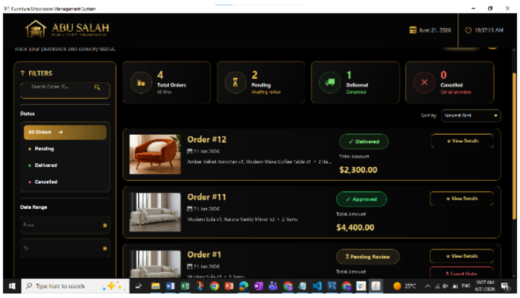


Figure 37: Customer Order Tracking After Delivery Completion

6.2 Test Scenario 2: Warehouse Response – Item Not Available

During this test, a customer submitted an order containing a product that was out of stock. The sales employee forwarded the order to the warehouse manager for inventory verification. The warehouse manager reviewed the available inventory levels and confirmed that the requested item was unavailable. The system automatically generated a stock verification response indicating that the requested quantity could not be fulfilled. The warehouse response was then displayed to the sales employee, allowing appropriate decision-making regarding the order.

Actual Result

- The inventory check correctly identified that the requested item had zero available units.

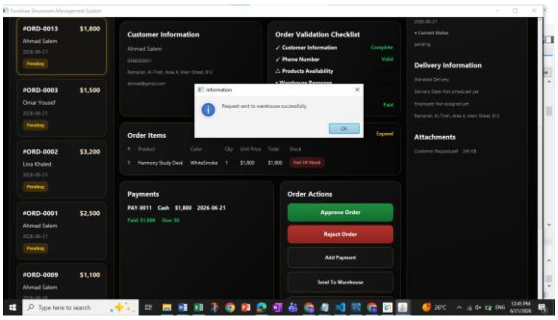
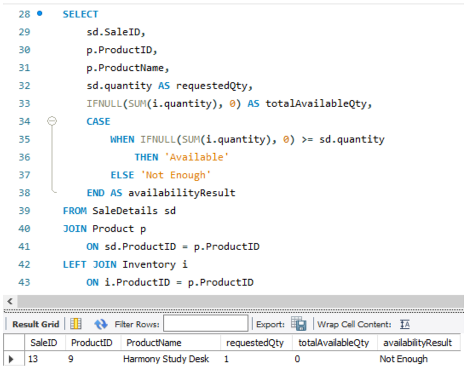


Figure 38:Order Sent to Warehouse for Stock Verification

- The warehouse response was successfully generated and displayed.



```sql
SELECT
    sd.SaleID,
    p.ProductID,
    p.ProductName,
    sd.quantity AS requestedQty,
    IFNULL(SUM(i.quantity), 0) AS totalAvailableQty,
    CASE
        WHEN IFNULL(SUM(i.quantity), 0) >= sd.quantity
            THEN 'Available'
        ELSE 'Not Enough'
    END AS availabilityResult
FROM SaleDetails sd
JOIN Product p
    ON sd.ProductID = p.ProductID
LEFT JOIN Inventory i
    ON i.ProductID = p.ProductID
```

SaleID	ProductID	ProductName	requestedQty	totalAvailableQty	availabilityResult
13	9	Harmony Study Desk	1	0	Not Enough

Figure 39 :Database Verification of Inventory Availability

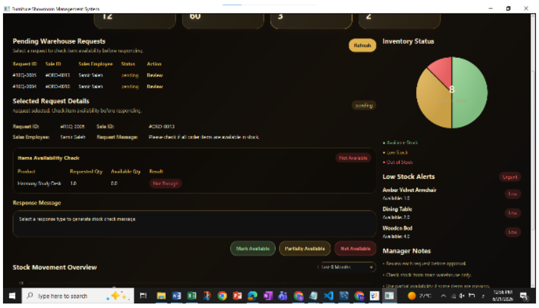


Figure 40 :Warehouse Manager Inventory Validation

- The sales employee received the warehouse verification result without errors.

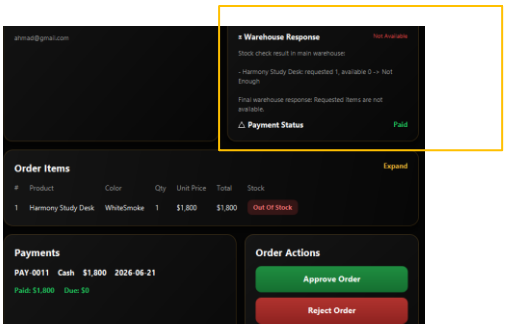


Figure 41 :Warehouse Response Returned to Sales Employee

- Customer View of Rejected Order.

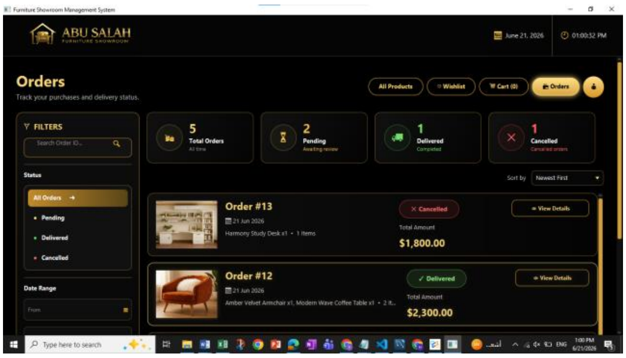


Figure 42:Customer View of Rejected Order

6.3 Test Scenario 3: Warehouse Response – Partially Available Items

Objective

To verify that the warehouse manager can identify orders where some requested items are available while others are unavailable, and that the partial availability response is correctly communicated to the sales employee.

In this scenario, the customer submitted an order containing multiple products. The sales employee forwarded the order to the warehouse manager for stock verification. During the inventory review, the warehouse manager found that some requested products were available in stock, while another product was unavailable. The system generated a Partially Available warehouse response and returned the result to the sales employee. The detailed response included the requested quantity, available quantity, and availability status for each item. This information enabled the sales employee to evaluate the order before taking further action.

➢ **Warehouse Inventory Availability Check**

The warehouse manager reviewed the requested order items and compared the requested quantities with the available inventory quantities. The system identified that some products were available while one product was out of stock.

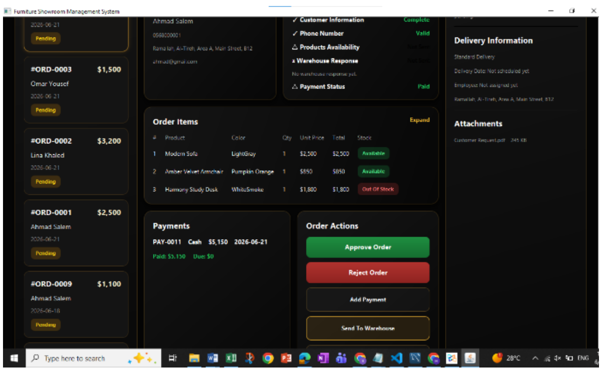


Figure 43: Warehouse Inventory Availability Check

➢ Warehouse Manager Submits Partial Availability Response

The warehouse manager selected the Partially Available response option and submitted the inventory verification result to the sales employee.

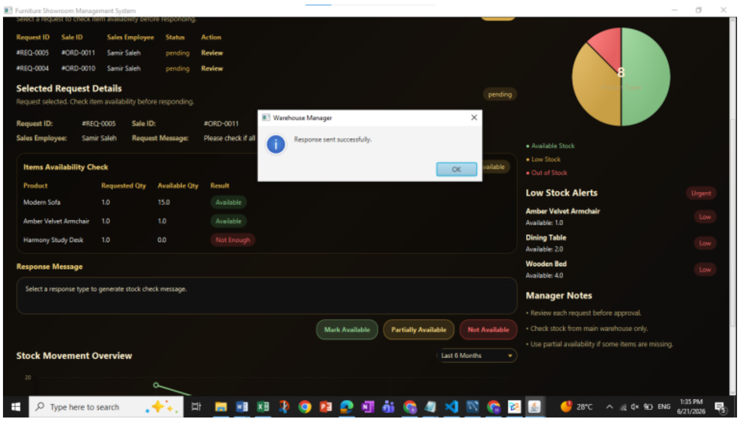


Figure 44: Warehouse Manager Submits Partial Availability Response

➢ Sales Employee Receives Partial Availability Response

The sales employee successfully received the warehouse response. The system clearly identified which products were available and which products were unavailable, allowing informed decision-making before order approval.

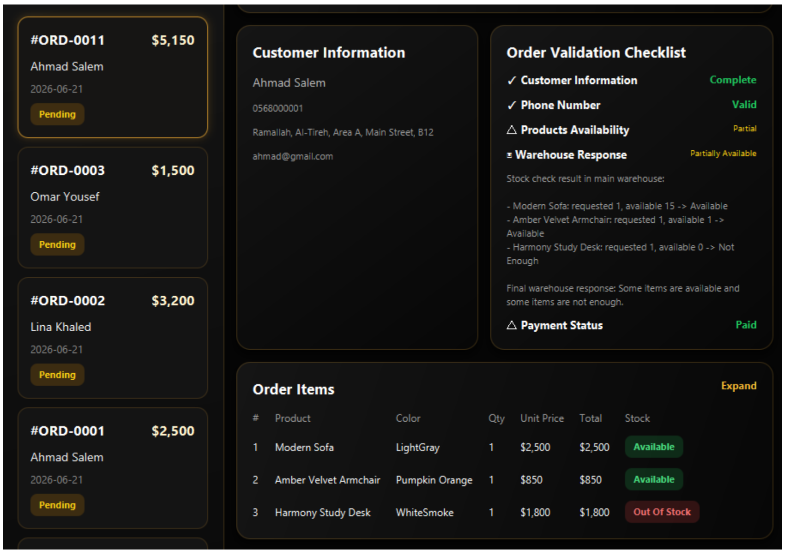


Figure 45:Sales Employee Receives Partial Availability Response

6.4 Test Scenario 4: Admin Product Management

This scenario verifies that the admin can manage product records through the Products module. The admin can view product details, check stock information, identify the related warehouse and supplier, and access product management actions such as update and delete. This confirms that the administrative product management interface is connected correctly with the database.

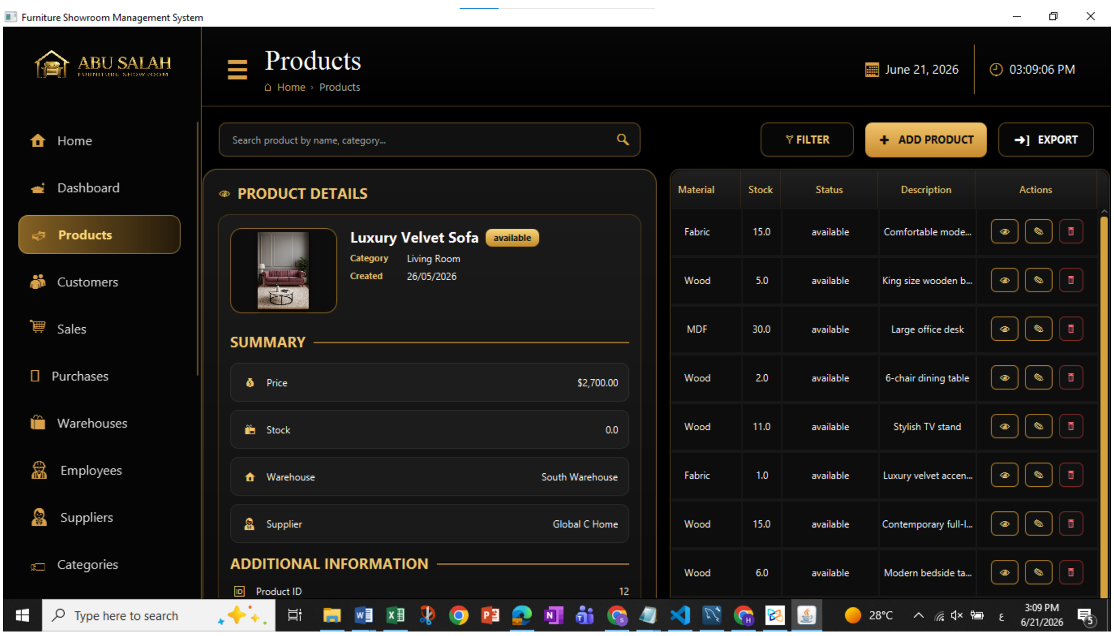


Figure 46:Admin Product Management and Product Details

Note: The presented screen represents one administrative module within the system. The administrator role is not limited to product management; it also includes managing customers, employees, suppliers, warehouses, inventory, categories, purchases, sales activities, and system reports. The Products module is shown as a representative example of the administrator's capabilities.